The Nostalgic Undertow: The Inevitability of Memory Obsolescence in Non-Stationary Landauer Regimes

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Abstract. A system remembers the world, and the world changes. Old memories gradually cease to predict anything, yet keeping them still costs energy. We call this growing fraction of useless but still-paid-for memory *informational nostalgia* (a formal quantity, not a psychological referent), and the irreducible rate at which it drags memory's efficiency toward zero the *undertow*. Our central result: the undertow is inevitable, by whatever route the system escapes — provided the environment drifts slowly and forgets its past. We close three conceivable escape routes, all dead ends. The first — track the environment more precisely, so tracking cost stops growing — fails: the optimal filter hits an irreducible limit on tracking a moving target, and the promised gain never appears. The second — refresh memory faster, washing out the stale before it accumulates — leaves, at any finite refresh rate, an irreducible residue of useless memory whose depth is the same for all such environments, set by the refresh rate alone, not the drift's form. The third — grow capacity faster than memory becomes obsolete — springs a different trap: no polynomial expansion helps, and the only regime that formally keeps memory useful, the exponential, is paid for by a diverging cost of expansion and storage, so efficiency still falls to zero. Each route is closed by a theorem; together they turn the undertow's inevitability from a model-specific observation into a structural closure of escape routes. All results are proved for mixing, slowly drifting environments in the linear-Gaussian (adiabatic) approximation.

Keywords: stochastic thermodynamics, Landauer's principle, non-stationary information, memory cost, Ornstein–Uhlenbeck process, predictive information, dynamic regret

1. Introduction

A learning system stores a memory of the world, but the world changes over time. A fine-tuned neural network, an adaptive controller, a living brain — each accumulates a model of an environment that drifts beneath it. The same conflict arises in every case: memory, once written, depreciates as a prediction but not as an obligation — holding each bit against thermal erasure is a continuous expenditure of free energy, whether or not it still predicts anything. Hence the question: is there an irreducible limit on the fraction of memory that no longer predicts anything yet is still paid for? Landauer's principle [1, 2] sets the price: holding a bit against erosion costs of order $k_B T \ln 2$ per correlation interval — this is the price of the continuous *maintenance* of written memory (in the sense of [3]), distinct from the classical erasure price, and paid continuously. The efficiency η_L — the ratio of the predictive information I_{pred} that the memory carries about the future of the environment to the total Landauer cost $N_{\max}$ of storing it (both inherited from [3] and introduced in § 2.1) — measures what fraction of this payment is recouped by predicting the current environment; our subject is its inevitable decline under drift.

The undertow metaphor. It is convenient to describe the totality of these effects as the *undertow*: the irreducible rate at which the efficiency of non-stationary memory is dragged toward zero under environmental drift — an underwater current that cannot be cancelled, only fought against with countervailing work. The central thesis of this work is that *the undertow is inevitable*: no conceivable escape route leads out from under it. The undertow has three measurable characteristics, each corresponding to one closed-off route: the *rate* (the speed at which the excess efficiency $\dot{\eta}_L^{\mathrm{excess}}$, the margin of η_L above the irreducible floor, decays — § 3, the precise-tracking route), the *depth* (the asymptotic level of nostalgia ν — § 4, the memory-refresh route), and the *attempt to escape* (the character of the growth of $|M_{\le t}|$ — § 5, the capacity-growth route).

The leading intuition: the budget channel. A single estimate clarifies why the undertow cannot be avoided "just by tracking more precisely". The predictive information about the current environment is bounded by the environment's alphabet, $I_{\mathrm{pred}} \le \ln k$ (where k is the number of distinguishable environment states) — no tracking strategy can raise the numerator of the efficiency above this ceiling, whereas the Landauer budget $N_{\max}(t)$ grows without bound for any working memory (at least linearly — the continuous payment for maintaining what is written against erosion). A bounded numerator divided by a diverging denominator gives $\eta_L(t) \to 0$ independently of the tracking strategy — this is the *budget channel* of vanishing, unconditional within the accounting postulates of § 2.1 (self-payment + stationary accounting) and requiring no assumptions about online learning. The efficiency thus vanishes through *two* channels: the *numerator channel* (nostalgia eats up the useful fraction of the prediction) and the *denominator channel* (the cost of growing memory diverges). The three routes below show that the numerator cannot be saved either: even where the denominator is held, nostalgia prevents the numerator from recouping the payment.

The inherited question (technical recap). The stochastic thermodynamics of memory describes the stationary regime well: the relation between the information retained about the future of the environment (the predictive information — the measure that separates the predictable part of the past from noise [4, 5]) and the cost of holding it is established for both model and biological systems [6, 7, 8], with the central Still–Sivak–Bell–Crooks inequality [6]: non-predictive memory is thermodynamically expensive. But this picture is derived for stationary protocols, whereas real systems operate differently — the environment drifts on the same time scales on which memory is refreshed and capacity grows, and the "information versus dissipation" balance acquires a temporal structure. The companion work [3] extended the balance to the non-stationary regime: it introduced the efficiency $\eta_L(t)$, the cumulative budget $N_{\max}(t)$ with its decomposition into storage E_{store}, growth E_{grow} and operational dissipation E_{actual}, and — centrally — *informational nostalgia* $\nu(t)$ (a technical term, a relational property of the memory–environment pair, not a psychological referent): the fraction of memory held against erosion but no longer reducing the mismatch between model and environment. The main result is Lemma 2: under slow OU drift and a finite refresh rate, the nostalgic fraction asymptotically does not fall below $1-c$, where $c = \dot{M}_{\text{refresh}}\,\tau_E/|M_{\leq t}|$ is assembled from measurable parameters. The load-bearing part of Lemma 2 is not the arithmetic definition of c, but the proof that unrefreshed bits are *informationally* useless: their contribution to the predictive information vanishes by the data-processing inequality through the decorrelation of the environmental latent. The definitions of the apparatus are given in § 2; the statements of the theorems follow below.

This result left three questions, explicitly listed in § 8.5 of [3]. First, the bound $\liminf \nu(t) \geq 1 - c$ is proved *only* for OU: both load-bearing steps — the exponential decorrelation of the optimal statistic and the additivity of mutual information across independently refreshed coordinates — are established only for the OU parametrisation of the logits and are *not* proved for long memory or discontinuous drift (Remark 4, § S5.1 [3]). Second, the *rate* of approach to zero remained a conjecture: § S8.1 [3] conjectured the adiabatic asymptotics $\dot{\eta}_L^{\text{excess}}(t) \sim -K \ln(\lambda t)/(2Rt^2)$ ($\lambda = 1/\tau_E$, so that λt is time measured in environmental correlation intervals), drawing on class II [4], but the scan confirmed only the functional form, while the convergence $B \to K/2$ remained unproved (a monotone growth of $B/(K/2)$ from 0.54 to 0.85 without reaching the limit). Third, the construction of Lemma 2 relies on the boundedness of memory through $c < 1$; under superlinear growth $|M_{\leq t}| \propto t^{\alpha}$ the condition is trivial, but the picture of nostalgia changes, and the extension remained open.

Three escape routes from the undertow — and why all are dead ends. We prove three theorems, closing off three routes and settling the three open questions of § 8.5 [3]; the leading result is *inevitability*. For emphasis, the positive dead ends come first: a proved floor (Theorem 2, C2) and a numerically confirmed saturation of capacity growth (Theorem 3, C3); Theorem 1 (C1) yields a *no-go of realizability* (Proposition 1, § 3) — the adiabatic logarithm of the tracking cost is unattainable as an asymptotic slope. The negative status of C1 *strengthens* the thesis: the precise-tracking route is closed off precisely because the promised positive asymptotics does not exist.

- **Theorem 1 (§ 3, C1) — the precise-tracking route — a dead end (no-go of realizability, Proposition 1).** The adiabatic BNT logarithm of the tracking cost $(K/2)\ln(\lambda t)$ under genuine OU drift is realized *only transiently* (a concentration burst $t \lesssim \tau_{\text{conc}}$), after which L_{excess} reaches a plateau and the fitted $B/(K/2) \to 0$: convergence as an *asymptotic slope* is **structurally unattainable** for a learner with saturating memory (σ-invariance of the log-carrying window, Proposition 1; growing memory does not save it — the floor is the informational optimum of tracking in the linear-Gaussian approximation A1 [open for the softmax model, § 7], old observations are decorrelated from the current $\theta(t)$, § 5, § 8). The analytical value $B = K/2$ survives only as the limit identity $\epsilon_{\text{track}} \to 0$ in the Kalman approximation (a static anchor without drift — slope 0.990; a transient burst under drift — 0.936), but does NOT imply an attainable positive asymptotics. The conjecture of § S8.1 [3] is thereby *resolved negatively* (aspect (i)): the form with $K/2$ is confirmed analytically, but is not realized as an attainable asymptotic law; the rigorous carry-over to the operational η_L is open (§ 7).
- **Theorem 2 (§ 4, C2) — the memory-refresh route — a dead end (a proved floor).** The nostalgia floor $\liminf \nu(t) \geq 1 - c$ holds beyond OU under a *sufficient mixing condition* $\beta(\Delta t) \to 0$: for every drift process with stationary increments that mixes, an irreducible residue $1 - c$ remains, and the floor constant is invariant across the class — no acceleration of refresh, short of an infinite rate, cancels it (verified on three representatives — PSP, fOU with Hurst exponent H, multi-scale drift). The *depth* of the floor is set by the refresh rate through c (invariant across the class), the *undertow-rate exponent* by the mixing β (for fOU it depends on H, for PSP on the switching frequency). The depth of the floor is proved; the exponent $4(1 - H)$ for fOU is secondary, confirmed on a correct stationary process in the far tail (§ 6.2, the `fou_floor` series). The reverse direction ($\beta \not\to 0 \Rightarrow$ the floor is destroyed) is given only qualitatively and is open (§ 4.6, § 7). The OU Lemma 2 [3] emerges as the special case $H = 1/2$.
- **Theorem 3 (§ 5, C3) — the capacity-growth route — a dead end (numerically confirmed saturation).** *No finite polynomial threshold α_c exists*: the undertow saturates ($\nu \to 1$) at any finite capacity-growth exponent $|M_{\leq t}| \propto t^{\alpha}$. Only exponential growth is capable of holding the nostalgic fraction below unity — but even it does not lead out from under the undertow; instead it shifts the collapse into another channel: the diverging cost of expansion $E_{\text{grow}}(t)$, so that $\eta_L(t) \to 0$ through the denominator. The phase boundary runs between polynomial and exponential growth, and on neither side is the route open (confirmed numerically, § 6, the `superlinear_memory` series).

Roadmap of mechanisms. Although there are three escape routes, there are only *two* load-bearing mechanisms. The first is the *decorrelation of old memory* (mixing of the environment): one and the same mixing-induced floor sets both the tracking floor

(§ 3, old observations decorrelate from the current $\theta(t)$) and the nostalgia floor (§ 4, unrefreshed bits vanish by DPI). The second is the *diverging cost of growing memory* (the budget channel of § 5, the denominator $N_{\max}(t) \to \infty$). The three theorems are three projections of these two mechanisms; their interlocking (the numerator and denominator channels lock each other) is discussed in § 8.

Relation to [3]. This work is an extension, not a revision. All definitions of the apparatus ($\eta_L(t)$, $N_{\max}(t)$, $L_{\text{excess}}(t)$, $\nu(t)$, the OU parametrisation of the logits) are taken from [3] without redefinition (§ 2 — a minimal summary). The OU results enter as special cases: Lemma 2 is the special case of Theorem 2 at $H = 1/2$; aspect (i) of the conjecture of § S8.1 is *resolved* by Theorem 1 (the value $K/2$ is confirmed analytically, but there is no attainable asymptotic realization under drift — a no-go, Proposition 1), while the rigorous carry-over to the operational η_L (aspect (ii)) is open (§ 7). This is a resolution of an open question, not a refutation.

Anchoring of the title metaphor. "Memory obsolescence" in the title is operationalized through the collapse of the efficiency $\eta_L \to 0$ (the budget channel, unconditional within the accounting postulates of § 2.1, inheriting [3] § 2.2); operational nostalgia $\nu^{\text{op}} \to 1$ as a direct referent of "obsolescence" is open (§ 7 item iii, the reverse direction of DPI). The title term is anchored to the proved η_L, not to the unproved ν^{op}.

Structure of the work. § 2 — setup and summary; § 3 — Theorem 1; § 4 — Theorem 2; § 5 — Theorem 3; § 6 — plan of the numerical verification; § 7 — relation to non-stationary online learning, Kalman tracking and renewal theory; § 8 — relation to articles #1 and #2.

2. Setup and a brief summary of the non-stationary scale

This section fixes the minimal set of definitions from [3] sufficient for reading the theorems of § 3–5, and introduces the taxonomy of drift classes on which Theorem 2 is built. The definitions are not redefined — they are given in the form of a methodological clarification with explicit references.

§ 2.1. Efficiency and budget. The non-stationary Landauer efficiency is defined in [3] (10) as

$$\eta_L(t) = \frac{I_{\text{pred}}(t,\tau)}{N_{\max}(t)},$$

where the numerator $I_{\text{pred}}(t,\tau) = I(X_t; X_{t+\tau} \mid \hat{P}(t))$ is the one-step predictive information under the current estimate of the environment $\hat{P}(t)$ [3] (4), bounded above by $\ln k$ for an alphabet of k states; the denominator is the cumulative Landauer budget [3] (1)

$$N_{\max}(t) = \frac{1}{k_B T \ln 2} \left(\int_0^t \dot{E}^{\text{curr}}_{\text{actual}}(t')\, dt' + E_{\text{store}}(t) + E_{\text{grow}}(t) \right),$$

decomposing into the current operational dissipation, the cumulative cost of holding the accumulated memory $E_{\text{store}}(t)$ and the cost of expanding capacity $E_{\text{grow}}(t)$. The conditional status of the bound $\eta_L \le 1$ (the self-payment postulate + the stationary-accounting postulate) is inherited unchanged [3] § 2.2.

§ 2.2. Nostalgia. Nostalgia $\nu(t)$ is the fraction of held memory that no longer reduces the mismatch between model and environment; in the theoretical normalisation [3] (7)

$$\nu^{\text{theor}}(t) = 1 - \frac{I_{\text{pred}}(t,\tau)}{I^{\text{opt}}_{\text{pred}}(t,\tau)}, \qquad I^{\text{opt}}_{\text{pred}}(t,\tau) = I(X_t; X_{t+\tau} \mid \theta(t)),$$

where $\theta(t)$ is the true latent parameter of the environment. *Operational* nostalgia $\nu^{\text{op}}(t)$ is normalised not on the oracular $I^{\text{opt}}_{\text{pred}}$, but on the limit $I^{\text{op}}_{\text{pred}}$ attainable under the best estimate $\hat{P}(t)$. By DPI $I^{\text{op}}_{\text{pred}} \le I^{\text{opt}}_{\text{pred}}$, whence (for a common numerator I_{pred}) operational nostalgia *does not exceed* the theoretical one:

$$\nu^{\text{op}}(t) = 1 - \frac{I_{\text{pred}}(t,\tau)}{I^{\text{op}}_{\text{pred}}(t,\tau)} \;\le\; 1 - \frac{I_{\text{pred}}(t,\tau)}{I^{\text{opt}}_{\text{pred}}(t,\tau)} = \nu^{\text{theor}}(t).$$

Therefore the lower bound proved for ν^{theor} does *not* carry over automatically to ν^{op} as a lower bound — the direction of the inequality is reversed; the carry-over remains structural (§ 7, item (iii)). The starting point of the present work is **Lemma 2** [3] § 4.4: in the OU class of slow drift, under FIFO refresh and $c := \dot{M}_{\text{refresh}}\,\tau_E/|M_{\le t}| < 1$, one has $\liminf_{t\to\infty} \nu^{\text{theor}}(t) \ge 1 - c$. The load-bearing part of Lemma 2 is not the arithmetic definition of c, but the proof that the contribution of the unrefreshed fraction to the predictive information vanishes by the data-processing inequality (DPI) through the decorrelation of the OU latent; it is precisely the confinement of this mechanism to the OU class that we lift in § 4.

§ 2.3. Excess loss. The cumulative excess loss of an ideal Bayesian learner relative to the oracle — *how much worse the real learner predicts than an all-knowing reference (the "oracle") that knows the true* $\theta(s)$ — is defined [3] (S8.1) as

$$L_{\text{excess}}(t) = \sum_{s\le t} \Big[\ln p(X_s \mid X_{s-1}, \theta(s)) - \ln \hat{p}(X_s \mid X_{s-1}, \hat{\theta}(s))\Big]$$

is measured in nats and is counterfactual (it requires access to $\theta(s)$); in learning theory this is the *dynamic regret* against an oracle that knows the drifting $\theta(s)$. The nat$\leftrightarrow$bit conversion is inherited from [3] App. A; η_L^{excess} is dimensionless under consistent units. The conjecture of § S8.1 [3] conjectured $L_{\text{excess}}(t) \sim (K/2)\ln(\lambda t)$ (class II [4], $K = k(k-1)$) and, for linear $N_{\max}(t) \sim Rt$, $\dot{\eta}_L^{\text{excess}}(t) \sim -K\ln(\lambda t)/(2Rt^2)$ for the secondary efficiency $\eta_L^{\text{excess}}(t) := L_{\text{excess}}(t)/N_{\max}(t)$. The analytical value $K/2$ is derived in § 3; the numerical attainment of the limit is open (§ 6.1, § 7).

§ 2.4. OU parametrisation. The canonical class of slow drift [3] § 5.2: the transition matrix $P(t)$ is specified through the logits $\theta_{ij}(t) \in \mathbb{R}^K$, satisfying the OU equation

$$d\theta_{ij} = -\lambda(\theta_{ij} - \theta^*_{ij})\,dt + \sigma\,dW_{ij}, \tag{2.1}$$

with a characteristic drift time $\tau_E = 1/\lambda$, a stationary variance $\sigma_0^2 = \sigma^2/(2\lambda)$ and the recovery of $P(t)$ through a row-wise softmax. The coordinates W_{ij} are independent; the covariance is $\mathbb{E}[\theta_{ij}(t)\theta_{ij}(s)] = \sigma_0^2\, e^{-|t-s|/\tau_E}$.

§ 2.5. Definition of slow drift. Two distinct adiabaticity conditions [3] § 5.2, § S8.2 are separated: (i) *slow-driving* $\lambda\tau_{\mathrm{relax}} \ll 1$ — the characteristic drift time of the environment is much greater than the relaxation time of the chain on each slice; necessary for Lemma 2 and the quasi-stationary treatment of $\theta(t)$. (ii) *OU-concentration* $\epsilon := \sigma^2/\lambda = 2\sigma_0^2 \ll 1$ — of the order of the stationary variance of the logit (a dimensionless smallness parameter), small compared with the drift scale; the concentration parameter of the OU latent itself. Both conditions together define the *adiabatic OU limit*. The hierarchy of times in this limit is $\tau_{\mathrm{relax}} \ll \tau_E \asymp \tau_f^{\mathrm{forget}}$ — the chain relaxes on each slice much faster than the environment drifts ($\tau_E = 1/\lambda$), and the optimal filter forgets on the drift scale ($\tau_f^{\mathrm{forget}} \asymp \tau_E$). *(Separately from this forgetting scale, the concentration time $\tau_{conc} \asymp 1/g^\star \propto 1/\sigma^2$ is introduced — item 5b below, § S1.2 — the duration of the transient burst over which the logarithm of Theorem 1 accumulates; in the adiabatic limit $\tau_{conc} \gg \tau_f^{forget}$.)*

§ 2.5a. The tracking control parameter ϵ_{track}. The convergence to the adiabatic limit for the asymptotics of L_{excess} (Theorem 1) is controlled *not* by $\epsilon = \sigma^2/\lambda$ itself (the concentration of the OU latent), but by a separate dimensionless *tracking* parameter

$$\epsilon_{\mathrm{track}} := \frac{\mathcal{I}_{\mathrm{rate}}\,\sigma^2}{\lambda^2}, \tag{2.2}$$

where $\mathcal{I}_{\mathrm{rate}}$ is the rate of accumulation of Fisher information about θ (the Fisher rate, § S1.2; for a Gaussian observation with noise r: $\mathcal{I}_{\mathrm{rate}} = 1/r$, so that $\epsilon_{\mathrm{track}} = \sigma^2/(\lambda^2 r)$ is tied to a measurable quantity). It is precisely $\epsilon_{\mathrm{track}}$ that governs the expansion of the stationary tracking error Σ_∞ (§ S1.2) and the attainment of the limit $B \to K/2$. Physically: ϵ tells how *narrow* the latent itself is, and $\epsilon_{\mathrm{track}}$ how well the filter manages to *track* it; therefore, even for an arbitrarily narrow latent, a finite $\mathcal{I}_{\mathrm{rate}}$ leaves a finite lag ($\epsilon_{\mathrm{track}} = O(1)$). The distinction is essential: the earlier softmax/RM scan [3] § S8.3 realized a small ϵ but stayed at $\epsilon_{\mathrm{track}} = O(1)$ and demonstrated only the *direction* of convergence; the new Kalman scan of § 6.1 *attains* a small $\epsilon_{\mathrm{track}}$ (down to $3 \cdot 10^{-2}$), and even there the slope is not recovered ($B/(K/2) \to 0$) — the load-bearing evidence of structural unattainability (§ 6.1, § S4.1).

§ 2.5b. The steady-state gain $g^\star$ and the two time scales of the filter. Two derived quantities of the optimal filter will be needed (full derivation — § S1.2). (i) The *steady-state Kalman gain* $g^\star := \mathcal{I}_{\mathrm{rate}}\Sigma_\infty/(1 + \mathcal{I}_{\mathrm{rate}}\Sigma_\infty\,\Delta t)$ (everywhere below $\Delta t = 1$, as in the simulations); in the adiabatic limit $g^\star = \mathcal{I}_{\mathrm{rate}}\sigma^2/(2\lambda)\,(1 + O(\epsilon_{\mathrm{track}})) \propto \sigma^2$ — it vanishes together with the drift. It sets the *floor of the effective number of observations* $n_{\mathrm{eff}}^\star = (1 - g^\star)/g^\star \approx 1/g^\star$: the geometric weighting of the past by the factor $(1 - g^\star)$ does not let n_{eff} grow without bound. (ii) Hence two time scales. The *forgetting scale* $\tau_f^{\mathrm{forget}} \asymp \tau_E = 1/\lambda$ — the memory depth of the filter under drift. The *concentration time* $\tau_{\mathrm{conc}} \asymp n_{\mathrm{eff}}^\star \asymp 1/g^\star \propto 1/\sigma^2$ — the duration of the transient burst up to the floor;

it is precisely τ_{conc}, not τ_f^{forget}, that bounds the window of logarithmic growth (§ 3). In the adiabatic limit the scales diverge: $\tau_{\text{conc}}/\tau_f^{\text{forget}} \asymp 1/g^\star = n^\star_{\text{eff}} \to \infty$. This separation is the key to the structural unattainability of the slope (§ 3, § S1.5): the available log-carrying window has a **σ-invariant** logarithmic width $W = t_{\max}/t_{\min} \asymp (1/\sigma^2)/\tau_{\text{conc}} \asymp \mathcal{I}_{\text{rate}}/\lambda = 1/(\lambda r)$ (σ cancels between $\tau_{\text{conc}} \propto 1/\sigma^2$ and the ceiling $1/\sigma^2$), so the limit $\epsilon_{\text{track}} \to 0$ does *not* open the window — both boundaries stretch in the same proportion (the window can be widened only by growing $\mathcal{I}_{\text{rate}}/\lambda \to \infty$, which exits the slow-drift regime).

§ 2.6. Taxonomy of drift classes. Theorem 2 is formulated for the class of drift processes with stationary increments and controlled mixing, generalizing OU. We single out four representatives:

- **OU** — a Gaussian Markov process (2.1) with exponential autocorrelation; $\rho(s) = e^{-|s|/\tau_E}$.
- **PSP** (piecewise-stationary Poisson) — the logits $\theta(t)$ are piecewise-constant, the jump points forming a Poisson stream of intensity $\mu = 1/\tau_E$; on each interval of constancy θ is independently re-drawn from the stationary distribution. This is a Markov process with discontinuous trajectories; "drift with reset" [3] § 6.1.
- **fOU** (fractional OU) — the stationary Gaussian solution of the Langevin equation with fractional Brownian motion B^H as the forcing (Hurst exponent $H \in (0,1)$); at $H = 1/2$ it coincides with OU, and at $H > 1/2$ — long memory (power-law, not exponential, decay of the autocorrelation) [9, 10].
- **multi-scale** — a finite superposition of OU processes with a hierarchy of times $\tau_E^{(1)} \ll \cdots \ll \tau_E^{(m)}$; the autocorrelation is a sum of exponentials.

For all classes a *mixing coefficient* $\beta(s)$ is introduced — the rate of decay of the conditional mutual information between the states of the environment at times t and $t-s$. For OU $\beta(s) \sim e^{-2s/\tau_E}$ (for a Gaussian pair $\beta \approx \frac{1}{2}\rho^2$, the square of the correlation doubling the exponent; § S2.2), for PSP $\beta(s) \sim e^{-s/\tau_E}$ (the probability of no jump), for fOU $\beta(s) \sim s^{-4(1-H)}$ at $H > 1/2$ (power-law), and for multi-scale — the slowest component $e^{-2s/\tau_E^{(m)}}$. It is precisely $\beta(s)$ that governs the undertow-rate exponent, while the floor constant $1-c$ is governed by the memory-refresh rate. This separation is the main structural finding of § 4: the form of β affects only the *rate* of the dragging toward zero, not the *asymptotic level*.

§ 2.7. Why the extension of the class is non-trivial. The proof of Lemma 2 in the OU class [3] § 4.4, § S1.5 rests on two specifically OU properties: (i) the *spectral gap* of the OU semigroup (exponential decorrelation with an explicit constant); (ii) first-order *Gaussian Markovianity* (a pure DPI chain $M^{\text{stale}} \to \theta(s) \to \theta(t)$). Both are violated for long memory: fOU at $H > 1/2$ is *not* Markovian and has *no* spectral gap. Theorem 2 removes the gap: neither the gap nor Markovianity is needed — mixing (4.1) suffices, and it holds also for the non-Markovian fOU. This is *not* a refutation of Lemma 2: Markovianity in its proof is a *sufficient*, not a necessary, condition; Theorem 2 extends the domain, it does not correct an error.

3. The precise-tracking route: the adiabatic identity $B = K/2$ and its transient realisability (Theorem 1, C1)

This section closes the first open question of § 8.5 of [3]: in the linear-Gaussian (Kalman) approximation, under explicit assumptions (A1)–(A3), it derives the analytical *value* of the logarithmic coefficient $B = K/2$ as an identity in the limit $\epsilon_{\text{track}} \to 0$, and shows that realising $K/2$ as an attainable asymptotic slope is structurally unattainable for a learner with saturating memory (Proposition 1) — resolving aspect (i) of the conjecture of § S8.1 negatively.

1. Setup and intuition. The conjecture of § S8.1 [3] rested on the logarithmic asymptotics of the excess loss: for a regular parametric model of dimension K with a slowly drifting latent, the cumulative excess loss of the ideal learner grows as $(K/2)\ln(\text{time})$. The one-half-per-parameter coefficient is the asymptotics of stochastic complexity [11, 12, 13], whose predictive-information form (with the argument of the logarithm shifted by the drift scale) is class II of Bialek–Nemenman–Tishby (BNT) [4]; it recurs in MDL, BIC, PAC-Bayes, and online Newton-step regret [11, 13, 14, 15]. The parametric scan [3] § S8.3 confirmed the functional form $At+B\ln(\lambda t)+C$ ($R^2 \geq 0.9995$), but the estimate $B/(K/2)$ fell short of unity ($0.54 \to 0.85$). Theorem 1 settles the *analytical* side: in the linear-Gaussian approximation under (A1)–(A3) the logarithmic coefficient is *identically* $K/2$ — but this does not mean the limit is reached as a slope over a wide asymptotic window. It is useful to separate *two pieces of evidence* in favour of $K/2$: (i) the analytical identity $B = K/2$ (the limit $\epsilon_{\text{track}} \to 0$) is confirmed by the **static anchor** — a run at *zero* drift, where the excess loss must be $(K/2)\ln t$ (slope 0.990, § 6.1 exp. A); (ii) at *non-zero* OU drift the same logarithm appears **only transiently** — as a spike over the window $t < \tau_{\text{conc}}$ (slope 0.936, exp. B), after which the drift caps n_{eff} at the floor $1/g^\star$, L_{excess} reaches a plateau, and the fitted $B/(K/2)$ over a wide window settles toward 0 (over 10 values of ϵ_{track} the series fluctuates in $[-0.31; 0.16]$ around zero, rather than growing toward 1, exp. C). The static anchor testifies to the *value* of the coefficient, the transient spike to realisability over a finite window only: the first is proved here, the second is structurally unattainable for a learner with saturating memory (§ 6.1, § 7).

The key object is the ideal Bayesian learner that estimates the drifting latent $\theta(t)$ from a stream of observations. In the adiabatic OU limit the optimal posterior estimate asymptotically coincides with an *exponentially-weighted filter* (EWMA) whose forgetting on the scale $\tau_f^{\text{forget}} \asymp \tau_E$ is tuned to the drift — the Bayesian counterpart of the stationary Kalman–Bucy filter for a linear-Gaussian system with a hidden OU state [16, 17]. The tracking-error formulation *derives* $K/2$ from the tracking dynamics rather than postulating it from an external result.

Theorem 1 (adiabatic asymptotics, OU). *Let the environment be given by the OU parametrization of the logits (2.1) in the adiabatic OU limit ($\lambda\tau_{relax} \ll 1$ and $\sigma^2/\lambda \ll 1$); let the learner be the ideal Bayesian one, asymptotically realised by an EWMA filter with optimal forgetting scale $\tau_f^{forget} \asymp \tau_E$. The proof rests on three explicit* *assumptions: (A1) the softmax observation model is linearised about the current estimate in the linear-Gaussian (EKF/Laplace) approximation, whose residual is controlled in the adiabatic limit $\sigma^2/\lambda \to 0$ (softmax regularity, [3] Remark 6); (A2) the adiabatic order of limits — first $\epsilon := \sigma^2/\lambda \to 0$ at fixed λt, or the transient window $1 \ll \lambda t \lesssim \lambda\tau_{conc} \asymp 1/g^\star$ (the limits $t \to \infty$ and $\epsilon \to 0$ do not commute); (A3) the effective number of informative observations concentrates as $n_{eff} \asymp \lambda t$ (a Bayesian concentration assumption of class II [4];* heuristically motivated* by analogy with Bartlett's effective sample size $n_{\text{eff}} = n/(1 + 2\sum_s \rho(s))$ [18] — a formal justification for *tracking* a drifting latent, as opposed to estimating a *mean*, is open, § 7). Then the cumulative excess loss is two-regime. **(a) Transient window** $1 \ll \lambda t \lesssim \lambda\tau_{\textbf{conc}} \asymp 1/g^\star$ (the posterior-concentration spike): logarithmic growth*

$$L_{\text{excess}}(t) = A \cdot t + \frac{K}{2}\ln(\lambda t) + C + o(1), \qquad 1 \ll \lambda t \lesssim \lambda\tau_{\text{conc}} \asymp 1/g^\star \text{ (A2)}. \tag{3.1}$$

***(b) Asymptotics** $t \gg \tau_{conc}$ (steady drift): the true $t \to \infty$ form is line-plus-plateau*

$$L_{\text{excess}}(t) \asymp A \cdot t + \frac{K}{2}\ln\frac{1}{g^\star} + o(t), \qquad t \gg \tau_{\text{conc}}, \tag{3.1$'$}$$

— *the logarithm* $(K/2)\ln(\lambda t)$ *does* not* continue indefinitely but saturates at a plateau of height $(K/2)\ln(1/g^\star)$ (the steady-state $g^\star$ caps n_{eff} at the floor $n^\star_{\text{eff}} = (1-g^\star)/g^\star \approx 1/g^\star$; Proposition 1, § S1.4). Both forms (3.1) and (3.1′) are equal parts of the theorem: (3.1) lives only on the transient window, (3.1′) is the genuine $t \to \infty$ regime.*

where $K = k(k-1)$ is the dimension of the logits, $A = A(\sigma^2/\lambda) \geq 0$ is the linear coefficient of the stationary tracking error, vanishing in the limit $\sigma^2/\lambda \to 0$, and the coefficient of the logarithmic term $B := K/2$ is independent of σ. As a consequence, under linear budget growth $N_{max}(t) \sim Rt$ the secondary efficiency $\eta_L^{excess}(t) = L_{excess}(t)/N_{max}(t)$ satisfies, on the transient window,

$$\dot{\eta}_L^{\text{excess}}(t) \sim -\frac{K\ln(\lambda t)}{2Rt^2}, \qquad 1 \ll \lambda t \lesssim \lambda\tau_{\text{conc}}. \tag{3.2}$$

Formula (3.2) holds on the transient window $1 \ll \lambda t \lesssim \lambda\tau_{conc}$; for $t \gg \tau_{conc}$ the rate of the undertow is set by the plateau $(K/2)\ln(1/g^\star)/N_{\max}$ (Proposition 1).

At finite amplitude $\epsilon := \sigma^2/\lambda$ the coefficient observed in the fit has the form $B(\epsilon) = (K/2)(1 - g(\epsilon))$ with $g(\epsilon) \to 0$ as $\epsilon \to 0$ — a finite-window bias of the estimate, one cause of the shortfall $B/(K/2) < 1$. The analytical identity $B = K/2$ pertains to the limit $\epsilon \to 0$ and does not assert that the limit is reached as a slope (the mechanism is given in the sketch below and in Proposition 1).

Leading intuition for the mechanism: the no-go rests on the strict finiteness of Σ_∞, not on the heuristic A3. The *qualitative* no-go (the logarithm is transient, L_{excess} reaches a line-plus-plateau, the fitted slope $B \to 0$ over a wide window) already follows from the **strict finiteness of the stationary tracking error** $\Sigma_\infty < \infty$ — the positive root of the algebraic Riccati equation (§ S1.2), a *rigorous* rather than heuristic result. The chain: $\Sigma_\infty < \infty \Rightarrow$ the posterior variance about the *current* $\theta(t)$ is bounded below by $\Sigma_\infty > 0$ for all t (the MMSE optimum, not exceeded by any memory, § S1.6) $\Rightarrow n_{\text{eff}}$ saturates at the floor $n^\star_{\text{eff}} = (1 - g^\star)/g^\star < \infty \Rightarrow$ the logarithmic integral is cut off at τ_{conc}, and the logarithm is transient (plateau). No link uses $n_{\text{eff}} \asymp \lambda s$. The heuristic A3 is needed **only** for two quantitative details *inside* the transient: (i) the *value* of the coefficient $K/2$ of the logarithmic spike, and (ii) the precise *scale* $\tau_{\text{conc}} \asymp 1/g^\star$. Transience as such, the existence of the plateau, and the vanishing of the asymptotic slope do *not* depend on A3 — they rest on the rigorous Riccati ($\Sigma_\infty < \infty$) plus the floor optimum of § S1.6: even if A3 is quantitatively wrong, the no-go (transience) survives, and only the numerical coefficient inside the spike changes. A3 fixes the *value* $K/2$ of the transient log, not the *fact* of transience. This rigorous scaffold ($\Sigma_\infty < \infty \Rightarrow$ saturation of $n_{\text{eff}} \Rightarrow$ transient log $\Rightarrow$ plateau) is primary; the sketch below merely refines, via A3, *where inside the transient the value $K/2$ comes from.*

2. Sketch of the mechanism. The full derivation is § S1 (Steps A–D: quadratic expansion S1.3, Riccati equation S1.2, integration of $K/2$ over the transient window S1.4, finite-amplitude correction S1.5). In compressed form: the instantaneous increment of the excess loss is a quadratic form of the tracking error in the Fisher metric (the KL expansion of Clarke–Barron [12, 11]). The error splits into a *stationary* part $\Sigma_\infty = \sigma^2/(2\lambda)(1 + O(\epsilon_{\text{track}}))$ (the root of the algebraic Riccati equation, § S1.2), giving the linear term $A\,t$ with $A \propto \sigma^2/\lambda \to 0$, and a *transient* part. The logarithm is born from the transient part: over the window $s \lesssim \tau_{\text{conc}}$ the posterior variance per coordinate falls as $1/(\mathcal{I}_{ij}\, n_{\text{eff}})$ with $n_{\text{eff}} \asymp \lambda s$ — observations older than τ_E decorrelate from $\theta(t)$, which replaces $\ln s$ by $\ln(\lambda s)$ (the class-II BNT shift [4] § VI), while the cancellation of Fisher factors leaves $K/2$ depending only on *dimension.* **Critically:** $n_{\text{eff}} \asymp \lambda s$ does not continue to $t \to \infty$ — under steady drift ($s \gtrsim \tau_{\text{conc}}$) the Kalman gain $g^\star \propto \sigma^2$ saturates n_{eff} at the floor $1/g^\star$, the integral is cut off at τ_{conc}, and the logarithm turns into a **plateau** $(K/2)\ln(1/g^\star)$; the premise $n_{\text{eff}} \asymp \lambda s$ and the saturation at the floor pertain to *different windows* of one integral, and do not contradict. The structural unattainability of the slope (via the σ-invariance of the log-carrying window) is unfolded in Proposition 1. □

3. What is proved and what is not. Theorem 1 establishes the *analytical* value of the coefficient (aspect (i) of § S8.1 [3]): $B = K/2$ in the linear-Gaussian approximation under (A1)–(A3). What it does *not* give: (0) the *numerical attainment* of the limit as a slope — the check of § 6.1 establishes that the logarithm $K/2$ is realised only transiently ($t \lesssim \tau_{\text{conc}}$), while over a wide window $\hat{B}/(K/2) \to 0$; the convergence is **structurally unattainable for a learner with saturating memory** ($\tau_{\text{conc}} \asymp 1/g^\star$ and the adiabatic ceiling $1/\sigma^2$ are both $\propto 1/\sigma^2$ and do not separate; growing memory does not save it — the floor is a lower bound across all architectures, Proposition 1, § S1.6), even though the static anchor confirms the value $K/2$ (0.990). Structural limitations remain: (i) the result pertains to the *secondary* η_L^{excess} on the counterfactual

L_{excess} (sample-level, see point 6), not to the operationally measurable $\eta_L(t)$ (10) — the link is structural, not identical; (ii) the realisability of the Bayesian update by an EWMA filter assumes $\tau_f \asymp \tau_E$; (iii) the derivation of $K/2$ rests on the stationary Riccati, the heuristic $n_{\text{eff}} \asymp \lambda t$ (A3), and softmax linearisation (A1) — a rigorous transient non-stationary filter and control of the linearisation residual are open (§ 7); for suboptimal learners (Robbins–Monro with $1/\sqrt{t}$ step, § S8.3) an additional bias appears over long horizons — an artefact of the step's suboptimality, not a property of the BNT effect (the negative control of § S8.3 [3]).

The negative content of Theorem 1 — the structural unattainability of the slope — we single out as a named statement (the main independent contribution of C1, rather than the identity $B = K/2$).

Proposition 1 (no-go of realizability of the adiabatic slope). *Under genuine OU drift ($\sigma > 0$) the BNT logarithm $(K/2)\ln(\lambda t)$ is realised only transiently — over the concentration window $t \lesssim \tau_{conc}$ — after which the steady-state Kalman gain $g^\star \propto \sigma^2$ caps the effective number of observations at the floor $n^\star_{\mathit{eff}} = (1 - g^\star)/g^\star \approx 1/g^\star$, and L_{excess} reaches a constant plateau of height $\sim (K/2)\ln(1/g^\star)$. Consequently, for a learner with saturating memory (the optimal stationary filter) the numerical convergence of the fitted coefficient $\hat{B}$ to $K/2$ as an asymptotic slope over a wide window ($t \gg \tau_{conc}$) is unattainable for any ϵ_{track}: the logarithmic width of the available log-carrying window $W = t_{\max}/t_{\min} \asymp (1/\sigma^2)/\tau_{conc} \asymp \mathcal{I}_{rate}/\lambda = 1/(\lambda r)$ is **σ-invariant** (σ cancels between $\tau_{conc} \propto 1/\sigma^2$ and the ceiling $1/\sigma^2$), so the adiabatic limit $\epsilon_{track} \to 0$ does* not* open the window — both bounds stretch in the same proportion, and the ratio is fixed. The inseparability is a consequence of the *adiabatic constraint* (a fixed Fisher rate per correlation interval of the environment), not of an arithmetic coincidence of the powers of σ: one can widen W only by growing $\mathcal{I}_{\text{rate}}/\lambda \to \infty$, which exits the slow-drift regime (§ S1.2). For any learner with growing memory the conclusion is the same: the floor $n^\star_{\text{eff}} = 1/g^\star$ is the informational optimum of tracking a drifting latent, exceeded by no memory architecture (see below). The proposition is conditional on (A1)–(A3) and the linear-Gaussian adiabatic approximation: it is proved in the adiabatic window ($\lambda\tau_{\text{relax}} \ll 1$, $\sigma^2/\lambda \ll 1$), and a rigorous version beyond it is open (§ 7 point iv).* This is a structural argument, not a defect of the experiment: the analytical identity $B = K/2$ in the limit $\epsilon_{\text{track}} \to 0$ and its unattainability as a slope at $\sigma > 0$ are *consistent* statements about different regimes, and it is precisely their joint validity that means the promised positive asymptotics of the tracking cost does not exist.

Optimality of the floor as a lower bound (not a postulate). Let us separate two statements: the *plateau of the logarithm* is a property of the *dynamics of a single filter* (n_{eff} saturates at the floor $1/g^\star$ for $t \gg \tau_{\text{conc}}$), whereas *floor = optimum* is an *informational lower bound across all memory architectures* (§ S1.6), explaining why the plateau cannot be circumvented even by growing memory. This is a provable bound in the linear-Gaussian window (A1): the Kalman posterior $(\hat{\theta}_t, \Sigma_t)$ is an *exact finite-dimensional sufficient statistic* for $\theta(t)$ given $Y_{1:t}$, and Σ_∞ is the Bayes-optimal (MMSE) tracking error [17, 16]. For *any* memory $S = f(Y_{1:t})$ the data-processing inequality gives $I(S;\theta(t)) \le I(Y_{1:t};\theta(t))$, and the optimal filter attains the bound; hence $n_{\text{eff}}(S) \le n^\star_{\text{eff}} = (1-g^\star)/g^\star$ for any architecture — the floor is the *maximum* of n_{eff}, not the limit of a particular memory. The decorrelation of old observations $\beta(\Delta t) \to 0$ (the mechanism of Theorem 2, C2) is an *illustrative parallel*, not the load-bearing argument; the load-bearing one is sufficiency + MMSE. **An essential caveat on generality:** the finite-dimensionality of the sufficient statistic rests on (A1); for a nonlinear/non-Gaussian model the posterior is not finite-dimensional, and the MMSE-optimality of the floor is not guaranteed — open (§ 7 point iv). Within (A1) the loophole is closed by the optimality of the floor; if memory is grown anyway, it runs into C3 (§ 5, § 8).

4. The role of Fisher information and universality. The substantive point (Step C, § S1.4) is the *cancellation of Fisher information*: the transient variance per coordinate decays as $1/(\mathcal{I}_{ij}\, n_{\text{eff}})$, while the quadratic form of the excess loss carries $\mathcal{I}_{ij}$ in the numerator; their product is independent of $\mathcal{I}_{ij}$, and coordinates of any curvature contribute equally 1/2. Hence $K/2$ depends only on the *dimension* of the parameter — the structural reason for the universality of class II, derived here without MDL/BIC as an external result.

5. Interpretation for the undertow. Theorem 1 fixes the *rate* of the undertow for the excess metric: η_L^{excess} decays to zero as $-\ln(\lambda t)/t^2$ — almost $1/t^2$ with a logarithmic correction. The drag is slow but irreducible: $\int^\infty |\dot\eta_L^{\text{excess}}|\, dt$ converges, but the *rate* of decay does not vanish at non-zero drift. The logarithmic factor is a signature of the dimension of the environment: the larger K, the steeper the undertow. Under linear budget growth $N_{\max} \sim Rt$ the undertow follows law (3.2); under exponential growth (§ 5) the denominator dominates, and the undertow accelerates to exponential — the link between Theorem 1 and Theorem 3.

6. Novelty of Theorem 1 and its level. The load-bearing mechanism is known *component by component*: the stationary Riccati and the Kalman–Bucy filter [16, 17], the quadratic KL expansion [12], the $\frac{1}{2}$-per-parameter coefficient of class II BNT [4] (also MDL/BIC/PAC-Bayes [11, 13, 14]), and the logarithmic tracking-regret from online convex optimization [19, 15]. New are three points absent from this literature. **(i) The asymptotic value of the coefficient in thermodynamically normalised form.** *The act* of normalisation $\eta_L^{\text{excess}} = L_{\text{excess}}/N_{\max}$ was introduced in [3] § S8.1, not here; what is new is the *asymptotic value* of the coefficient: $\dot\eta_L^{\text{excess}} \sim -K\ln(\lambda t)/(2Rt^2)$ with the derived $K/2$ and the scale $k_BT\ln 2$ per bit. The coupling "logarithmic regret with a *derived* $K/2 \leftrightarrow$ thermodynamic efficiency with the scale $k_BT\ln 2$" is present neither in BNT, nor in Kalman tracking, nor in tracking-regret: there the coefficient remained a conjecture. **(ii) The no-go of realizability itself (Proposition 1) — the main independent contribution of C1.** The *fact* of degradation of static log-regret under non-stationarity is known in the adaptive-regret literature [20, 21]; our contribution is the *structural irreducibility* specifically for tracking an OU latent: the inseparability of τ_{conc} and the adiabatic ceiling $1/\sigma^2$ (both $\propto 1/\sigma^2$), plus the thermodynamic normalisation of the plateau. BNT and Clarke–Barron give the logarithm for the *stationary* problem; we show *why precisely* OU drift kills it as an attainable slope, leaving only the spike and the plateau. **(iii) The transfer of the tracking-error mechanism to the mixing classes fOU and PSP** (§ 4) — beyond the Markovian OU. An essential caveat on level (consistency with [3]): Theorem 1 gives the asymptotics for the *sample-level* excess loss — this is *weaker* than the bit-level result of [3]; the transfer to the operational η_L remains structural, not identical (excess-loss = sample-level regret, not bit-level erosion of memory).

4. The memory-refresh route: universality of the nostalgia floor (Theorem 2, C2)

This section closes the second open question of § 8.5 of [3]: it carries the lower bound $\liminf \nu(t) \geq 1 - c$ beyond the OU class. The key observation: both load-bearing steps of Lemma 2 — decorrelation and additivity of mutual information — require not a specifically OU structure, but only (i) controlled mixing of the environment and (ii) stationarity of the latent's increments. We formalise the sufficient class and show that OU is its special case $H = 1/2$.

1. The class of processes. Call a drift process $\theta(t)$ *admissible* if it is Gaussian (or asymptotically Gaussian), wide-sense stationary, has stationary increments, and satisfies a *mixing condition*: there exists a non-increasing function $\beta(s) \to 0$ such that, for the conditional mutual information of the latent at time s with the *future trajectory* of the latent over the prediction horizon,

$$I\big(\theta(s);\, \theta_{[t,t+\tau]} \mid \mathcal{F}_{\mathrm{refr}}\big) \leq C \cdot \beta(t-s), \qquad \int_0^\infty \beta(s)\, ds < \infty, \tag{4.1}$$

where $\mathcal{F}_{\mathrm{refr}} := \sigma(M^{\mathrm{refr}}_{\leq t})$ is the σ-algebra generated by the refreshed fraction of memory $M^{\mathrm{refr}}_{\leq t}$ (throughout, $\mathcal{F}_{\mathrm{refr}}$ and conditioning on $M^{\mathrm{refr}}_{\leq t}$ are synonymous, § S2.0). The four classes of § 2.6 are admissible: OU, PSP and multi-scale with exponential β, and fOU with power-law $\beta(\Delta t) \sim \Delta t^{-4(1-H)}$, for which the convergence condition $\int_0^\infty \beta\, ds < \infty$ holds at $H < 3/4$ (the range $H \in [3/4, 1)$ is not covered by the rigorous theorem; see § 4.6 and § S2.2). Condition (4.1) is a direct generalization of the OU spectral gap [3, 22] to processes for which the gap may be absent but mixing (at least power-law) is present.

Theorem 2 (universality of the nostalgia floor). *Let the environment X_E be governed by an admissible drift process $\theta(t)$ (4.1) in the slow-drift regime. The slow-drift condition here is twofold and holds for all classes: (i) $\tau_{relax} \ll \tau_E$ — the chain relaxes to quasi-stationarity between significant changes of the latent (for PSP, between jumps, i.e. $\tau_{relax} \ll 1/\mu$), which justifies the quasi-stationary treatment of $\theta(t)$ on each slice; (ii) $\beta(\tau_E) \ll 1$ — mixing of the latent on the characteristic scale τ_E. Memory is refreshed by FIFO at finite rate $\dot{M}_{refresh}$; $c := \dot{M}_{refresh}\, \tau_E / |M_{\leq t}| < 1$ for all $t > T_0$; per-bit uniformity holds —* uniformity of the per-bit contribution*: under a factorised posterior and symmetric FIFO the limiting predictive contribution of a single bit does not exceed the average $I^{opt}_{pred}/|M_{\leq t}|$ (this premise is* stronger* than plain additivity of mutual information: additivity by itself does not forbid fresh bits from carrying a disproportionately large share; uniformity of the contribution additionally requires the symmetry of independent coordinates, which is precisely what additivity of MI over symmetric independent coordinates provides, [3] Remark 5; § S1.1). Then*

$$\liminf_{t\to\infty} \nu^{\mathrm{theor}}(t) \;\geq\; 1-c. \tag{4.2}$$

The floor constant $1-c$ *depends only on the memory-refresh rate and the characteristic mixing scale* τ_E*, and is independent of the form of* $\beta(\cdot)$*. The rate of approach to the floor is governed by* $\beta(\cdot)$*: the contribution of the stale fraction to the predictive information decays as* $\beta(\Delta t)$ *with the age* Δt *of the un-refreshed bit.*

Let us stress the asymmetry of the two parts of the result: the *depth* of the floor $(1-c)$ is proved strictly and is invariant to the form of $\beta(\cdot)$ (the load-bearing contribution of C2), whereas the *rate exponent* is secondary, depends on the microstructure of β, and is *asymptotic* (§ 6.2, § 7 point vi).

A normalisation caveat (stated locally, so that § 4 reads in isolation): bound (4.2) is proved for the *theoretical* normalisation ν^{theor} (normalisation by the oracle $I_{\mathrm{pred}}^{\mathrm{opt}}$). The transfer to the *operational* ν^{op} (normalisation by the attainable-under-estimate $I_{\mathrm{pred}}^{\mathrm{op}} \leq I_{\mathrm{pred}}^{\mathrm{opt}}$) is *open*: the direction of the DPI is reversed ($\nu^{\mathrm{op}} \leq \nu^{\mathrm{theor}}$, § 2.2), so the lower bound for ν^{theor} does *not* transfer automatically to ν^{op} as a lower bound (§ 2.2, § 7 point iii). In other words, (4.2) should not be read as $\nu^{\mathrm{op}} \geq 1-c$.

2. Sketch of the proof. The full derivation, with three sub-cases, is § S2. The structure follows [3] § 4.4, but the decorrelation step is carried out through the general mixing condition (4.1) rather than through the OU spectral gap.

The chain rule for mutual information gives a decomposition over the refreshed and stale fractions:

$$I(M_{\leq t}; X_E^{[t,t+\tau]}) = I(M_{\leq t}^{\mathrm{refr}}; X_E^{[t,t+\tau]}) + I(M_{\leq t}^{\mathrm{stale}}; X_E^{[t,t+\tau]} \mid M_{\leq t}^{\mathrm{refr}}).$$

The stale fraction $M_{\leq t}^{\mathrm{stale}}$ is a deterministic function of $\theta(s)$ ($s \leq t - \Delta t$). The Markov property of the drift is *not* used: two data-processing steps suffice — $M^{\mathrm{stale}} = f(\theta(s))$ and $X_E^{[t,t+\tau]} = g(\theta_{[t,t+\tau]})$ — giving the DPI bound

$$I(M_{\leq t}^{\mathrm{stale}}; X_E^{[t,t+\tau]} \mid M_{\leq t}^{\mathrm{refr}}) \leq I\big(\theta(s); \theta_{[t,t+\tau]} \mid M_{\leq t}^{\mathrm{refr}}\big) \leq C \cdot \beta(\Delta t),$$

where the last majorant is the mixing condition (4.1). It is precisely the abandonment of the Markov property that lets the argument work for the non-Markovian fOU at $H > 1/2$ (§ S2.0).

Under slow drift ($\beta(\tau_E) \ll 1$) and FIFO, averaging over the uniform distribution of ages makes the stale fraction's contribution to the predictive information vanish to accuracy $O(c)$, and the fraction of un-refreshed bits $\geq 1-c$ is converted — via per-bit uniformity — into the lower bound on nostalgia (4.2). In essence, the passage from the arithmetic fraction $1-c$ to the information-theoretic inequality requires exactly (4.1), not exponentiality; even the power-law mixing of fOU at $H > 1/2$ ensures $\beta(\Delta t) \to 0$, which suffices. $\square$

3. What governs the floor and the rate of the undertow. Theorem 2 separates two independent characteristics of the undertow.

- **The floor constant** $1 - c$ is governed by the *memory-refresh rate*: $c = \dot{M}_{\text{refresh}}\tau_E/|M_{\leq t}|$. *Under the mixing condition* $\beta(\Delta t) \to 0$ it is independent of the form of $\beta(\cdot)$ — for OU, PSP, fOU at any $H < 1$, and multi-scale at fixed τ_E the depth of the undertow is the same. This is not a tautology: c is assembled from measurable parameters before the microstructure is known, and (4.1) guarantees that the microstructure enters only the *rate*, not the *level* (no claim is made for non-mixing environments, § 4.6). *Novelty beyond ergodic folklore.* The implication "mixing $\Rightarrow$ decorrelation of old observations" is standard; the novelty of C2 lies elsewhere: (i) in the *translation* of decorrelation into a *thermodynamic* lower bound $\nu \geq 1 - c$ via Landauer normalisation (decorrelation by itself says nothing about the share of *paid* memory — this is supplied by the coupling with $N_{\max}$ and the definition of c); (ii) in the proof of *invariance of the depth* of the floor across the class under varying mixing rate (depth $\perp$ rate: $1 - c$ is independent of the form of β, the undertow exponent depends on it) — a separation present neither in OU Lemma 2 [3] nor in standard mixing theory.
- **The undertow-rate exponent** is governed by the *mixing* $\beta(\cdot)$. For OU $\beta(\Delta t) \sim e^{-2\Delta t/\tau_E}$ (the square of the Gaussian correlation); for PSP $\beta(\Delta t) \sim e^{-\Delta t/\tau_E}$ (the probability of no jump, $\tau_E = 1/\mu$); for fOU at $H > 1/2$, *power-law* $\beta(\Delta t) \sim \Delta t^{-4(1-H)}$ (larger H, slower approach to the floor, but the floor is the same $1 - c$); for multi-scale, the slowest component $\tau_E^{(m)}$. An essential caveat on the status of the exponent $4(1 - H)$ (structurally the same situation as transience in C1): it is an *asymptotic, far-tail* prediction (§ S2.2) — on an exact stationary fOU it is *reproduced* in the far tail ($\Delta \gtrsim 8\tau_E$: 1.20 at $H = 0.7$, ≈ 0.4 at $H = 0.9$, § 6.2, § S4.2), but in the *near* window $[\tau_E, 8\tau_E]$ it is *not isolated* (the OU "knee" dominates, the sample ACF at large lags is biased toward zero), and is read off the exact covariance rather than the sample ACF. The status of the exponent does *not* undermine the universality of the floor: the load-bearing contribution of C2 is the *depth* $(1 - c)$ (proved, invariant), and the *rate exponent* is secondary.

4. OU as a special case. Lemma 2 of [3] is a special case of Theorem 2 at $H = 1/2$ (fOU degenerates into OU) with $\beta(s) = e^{-2s/\tau_E}$: the specifics of OU (spectral gap, exponential mixing) were a *sufficient* but not a *necessary* condition for the floor; the necessary one is merely mixing (4.1) and stationarity of the increments.

5. Transfer of additivity and per-bit uniformity. The second load-bearing step of Lemma 2 is per-bit uniformity, the additivity of mutual information over independently refreshed coordinates [3] Remark 5. It carries beyond OU unchanged: additivity requires only *statistical independence* of the coordinates, not OU structure. For fOU with (i, j)-independent fractional forcings, for PSP with independent re-sampling at reset, and for multi-scale, the coordinates are independent by construction, and additivity holds with constant $1+o(1)$ (§ S2.0). *The same softmax caveat as in C1 (assumption A1).* Per-bit uniformity is exact for the factorisation of the *latent* θ_{ij}; the observable $P(t)$ is built by row-wise softmax, which couples the coordinates of a row, so the posterior over the observations does not factorise *exactly* — as in § 3, where the softmax linearisation is delegated to A1. The coupling introduces an $O(\epsilon)$ correction and does *not change the sign* of the bound: a correlated softmax posterior only *raises* the floor ν (positive correlation of a row's fresh bits reduces their joint contribution), it does not break through it. Rigorous control of the softmax correction is open (§ 7 point iv); in the adiabatic limit it is absorbed into $(1+o(1))$.

6. The boundary of the class. Theorem 2 asserts the *sufficiency* of mixing: $\beta(\Delta t) \to 0 \Rightarrow$ the floor $1-c$. The converse — that the *absence* of mixing ($\beta \not\to 0$) *destroys* the floor — is given only qualitatively, not as a theorem: for processes with divergent autocorrelation the stale fraction *may* retain a finite contribution, and the floor is not guaranteed, but we provide no rigorous proof of destruction. Condition (4.1) delineates the boundary of universality: the undertow is inevitable *at least* when the environment mixes; the *necessity* ($\beta \not\to 0 \Rightarrow$ no floor) is an open conjecture, not a theorem (§ 7). To avoid false attribution: the Besbes–Gur–Zeevi argument (§ 7 point 1a) on the impossibility of sublinear dynamic regret pertains to the *excess-loss metric* (C1), *not* directly to the ν-floor of C2; the link is structural, not identical (§ 7 point iii). A separate boundary is long-memory fOU at $H \in [3/4, 1)$: here $\beta(\Delta t) \to 0$ pointwise, but $\int_0^\infty \beta\, ds$ diverges, and the cumulative contribution of the stale fraction is not majorised by the pointwise decay; the rigorous Theorem 2 does *not* cover this range (§ S2.2), and the floor $1-c$ there is a conjecture with numerical support. At the boundary $H = 3/4$ ($4(1-H) = 1$) the divergence of $\int_0^\infty \beta\, ds$ is only *logarithmic* ($\beta \sim 1/\Delta t$), milder than the power-law one at $H > 3/4$ ($\int^T \beta \sim T^{4H-3}$); this makes $H = 3/4$ the most accessible target for a future argument. At $H = 1$ fOU degenerates into a process with conserved memory, and the theorem obviously yields no floor.

7. Connection with bet-hedging. Lemma 2 [3] § 4.4 has a structural predecessor in population biology — the Kussell–Leibler inequality for bet-hedging in fluctuating environments [23], where the value of memory about the environment is bounded by its correlation horizon: memory is beneficial only within it, and its retention beyond it does not pay off (there is a maximal cost at which memory is useful). Theorem 2 strengthens the parallel: the universality of the floor across the mixing class means that the fitness cost of stale information is independent of the *spectral* structure of the fluctuations (white noise, coloured, long-memory) — depending only on the ratio of refresh and drift rates. This predicts the robustness of bet-hedging strategies to the form of the environment's autocorrelation, consistent with the empirical universality of diapause and spore-forming mechanisms in environments of differing spectral structure [24, 25].

5. The capacity-growth route: super-linear memory growth and the escape phase boundary (Theorem 3, C3)

This section closes the third open question of § 8.5 [3]: the regime of super-linear memory growth $|M_{\le t}| \propto t^\alpha$, $\alpha \ge 1$ — can sufficiently fast capacity growth let the system *outrun* the undertow? We show that no finite polynomial threshold α_c exists (the undertow saturates at any finite α), that escape requires exponential capacity growth, and we establish its thermodynamic cost.

1. Setup. Let capacity grow as $|M_{\le t}| = M_0\, t^\alpha$, $\alpha \ge 0$, and the refresh rate as $\dot M_{\text{refresh}}(t) = \alpha M_0 t^{\alpha-1} + \rho_0$ (ρ_0, the background rate). Then the constant $c(t)$ of Lemma 2 becomes time-dependent:

$$c(t) = \frac{\dot M_{\text{refresh}}(t)\cdot \tau_E}{|M_{\le t}|} = \frac{(\alpha M_0 t^{\alpha-1} + \rho_0)\tau_E}{M_0 t^\alpha} = \alpha\tau_E\, t^{-1} + \frac{\rho_0\tau_E}{M_0} t^{-\alpha}. \tag{5.1}$$

The question of escape reduces to the asymptotics of $c(t)$: if $c(t) \to 0$, the floor $1-c \to 1$ and nostalgia saturates ($\nu \to 1$, the undertow wins); if $c(t)$ stays bounded away from zero, nostalgia is held below 1 (escape).

2. The phase boundary. From (5.1), both terms ($\alpha\tau_E/t$ and $(\rho_0\tau_E/M_0)t^{-\alpha}$) decay at $\alpha > 0$, so at first sight $c(t) \to 0$. But this ignores the *prediction horizon and the age distribution*: under FIFO the fraction of bits refreshed within a correlation interval τ_E is the ratio of the *increment* of capacity over τ_E to the total capacity. The quantity governing nostalgia is the *fresh-bit fraction*

$$\phi(t) := \frac{|M_{\le t}| - |M_{\le t-\tau_E}|}{|M_{\le t}|} = 1 - \left(\frac{t-\tau_E}{t}\right)^\alpha \approx \frac{\alpha\tau_E}{t} \quad (\alpha = O(1),\ t \gg \tau_E). \tag{5.2}$$

(Connection with (5.1): the leading term $\phi(t) \sim \alpha\tau_E/t$ coincides with the leading term of $c(t)$; the background ρ_0-term is sub-leading at $\alpha \ge 1$, so the governing quantity is $\phi(t)$.) The fresh fraction $\phi(t) \sim \alpha\tau_E/t \to 0$ at *any* finite α: pure capacity growth does not overcome the undertow, because the *fraction* of fresh relative to accumulated tends to zero. Escape requires growth not of capacity but of the *per-bit refresh rate* — the fresh-bit fraction must be bounded away from zero. Formally:

Theorem 3 (phase boundary of super-linear memory). *Let the memory capacity grow as $|M_{\le t}| \propto t^\alpha$ and let the refresh protocol update bits with age priority (FIFO). Introduce an effective refresh exponent: the fraction of bits refreshed over the last interval τ_E behaves as $\phi(t) \sim t^{-\gamma}$ with $\gamma = \gamma(\alpha, \text{protocol})$. Then there is a critical exponent $\gamma_c = 0$ such that:*

- *at $\gamma > 0$ (including pure capacity growth at any finite α, for which $\phi(t) \to 0$) — a saturating undertow: $\liminf \nu^{\text{theor}}(t) \to 1$ (in the normalisation sense), and independently $\eta_L(t) \to 0$ through the divergent denominator $N_{\max}$ (point 4). The saturating (normalisation) part inherits the admissible mixing class of Theorem 2 ($\int \beta < \infty$; for fOU $H < 3/4$; the range $H \ge 3/4$ is open, § 4.6) — it is a pointwise* *application of S2.0 (§ 5 point 5a, § S3.2); the budget channel ($\eta_L \to 0$ through the denominator) does* not* depend on mixing and holds for any growing memory unconditionally (within the accounting postulates of § 2.1);*

- *at $\gamma = 0$ (a constant refresh fraction, attainable only under exponential capacity growth $|M_{\le t}| \propto e^{\kappa t}$ with $\kappa\tau_E = O(1)$, or under redistribution of refresh proportionally to capacity) — escape:* $\liminf \nu(t) \le 1 - c_\infty$ *with finite $c_\infty > 0$, and the nostalgic fraction stays bounded away from* 1.

A critical regime α_c for polynomial growth is absent in the proper sense: no finite polynomial exponent α yields escape; the boundary between saturation and escape runs between polynomial ($\gamma > 0$) and exponential ($\gamma = 0$) capacity growth. Equivalently, for polynomial memory escape requires growth of the per-bit refresh rate, $\dot{M}_{refresh}/|M_{\le t}| \not\to 0$, which is incompatible with finite $\dot{M}_{refresh}$ at $|M_{\le t}| \to \infty$.

3. Sketch of the proof. The full derivation is § S3. From (5.2) the fresh fraction $\phi(t) \sim \alpha\tau_E/t \to 0$ at any finite α. A pointwise version of step S2.0 with $\phi(t)$ in place of the constant c (§ S3.2): since ϕ varies adiabatically relative to the window τ_E, the DPI majorant is applied on each window τ_E with locally-constant $\phi(t)$ (under per-bit uniformity and FIFO), giving $\nu(t) \ge 1 - \phi(t)(1 + o(1))$ and $\liminf \nu(t) \to 1$. This *adiabatic carry-over* of the S2.0 majorant (derived for constant c) to a slowly-varying $\phi(t)$, with $O(\tau_E/t)$ control of non-constancy, is a separate new step of C3 (§ S3.2; § 5 point 5a), not reducible to C2. The carry-over inherits the mixing hypothesis of Theorem 2 ((4.1), $\int \beta < \infty$; fOU $H < 3/4$; $H \ge 3/4$ open, § 4.6); the budget channel (point 4) does not depend on mixing. Escape is possible only when $\phi(t)$ is bounded away from zero — that is, under multiplicative (exponential) capacity growth on the scale τ_E. □

4. The thermodynamic cost of escape. Escape is not free. The cost of memory expansion $E_{\text{grow}}(t)$ [3] (1) is the work to create new capacity, bounded below by $k_BT \ln 2$ per bit. For the exponential capacity growth required for escape,

$$\dot{E}_{\text{grow}}(t) \ge k_BT \ln 2 \cdot \frac{d}{dt}|M_{\le t}| = k_BT \ln 2 \cdot \kappa M_0 e^{\kappa t}, \tag{5.3}$$

that is, the instantaneous power of expansion grows *exponentially*, and the cumulative $E_{\text{grow}}(t) \propto e^{\kappa t}$ diverges super-polynomially. **The load-bearing line of the derivation $\eta_L \to 0$ rests on E_{grow} alone** (5.3): already $N_{\max}(t) \ge E_{\text{grow}}(t)/(k_BT \ln 2) \ge M_0(e^{\kappa t} - 1) \to \infty$ with bounded numerator $I_{\text{pred}} \le \ln k$ gives collapse; $\tau_{1/2}$ does *not* enter the result. The storage cost E_{store} below is an *a-fortiori strengthening remark*, not a necessary parameter: under exponential capacity the *storage* cost also diverges, since holding $\propto e^{\kappa t}$ bits against erosion requires power (per-bit power, [3] (2))

$$\dot{E}_{\text{store}}(t) \gtrsim \frac{k_BT \ln 2}{\tau_{1/2}} \cdot |M_{\le t}| \propto e^{\kappa t}, \tag{5.4}$$

where $\tau_{1/2}$ is the half-erosion time of a bit (a power flow, not energy: the factor $1/\tau_{1/2}$ restores the dimension). Here a *weakened* naive bound (2) is used: the exact per-bit bound [3] (2) contains a tolerance prefactor $\ln 2/(2\varepsilon) \geq 1$, which we omit — the exact bound is only *larger*, so the conclusion holds *a fortiori*. The flows E_{grow} (creation of capacity) and E_{store} (its retention) are physically distinct works [3] § 2.2, accounted for separately without double-counting; the denominator is dominated by their sum, both $\propto e^{\kappa t}$. This violates the linear-budget premise $N_{\max} \sim Rt$ of Theorem 1: although nostalgia is held below 1, $\eta_L = I_{\text{pred}}/N_{\max}$ collapses to zero — not because of the numerator, but because of the exponential denominator.

Two distinct channels of vanishing η_L — and which is operational. (i) *The budget channel (denominator) is operationally reliable.* For any growing memory $N_{\max}(t) \to \infty$ (at least linearly $\sim Rt$, and under growing capacity $\propto t^{\alpha}$ via E_{store} or $\propto e^{\kappa t}$), while the numerator $I_{\text{pred}} \leq \ln k$ is bounded by the alphabet. Hence $\eta_L \to 0$ is *operationally guaranteed by the budget channel* — regardless of whether I_{pred} reaches a plateau or decays; the channel is unconditional (within the accounting postulates of § 2.1). (ii) *The normalisation channel (nostalgia ν) is a normalisation-theoretic statement.* The saturation $\nu^{\text{theor}} \to 1$ (S3.2) is the vanishing of the *ratio* $I_{\text{pred}}/I_{\text{pred}}^{\text{opt}}$, not of the absolute I_{pred} (which may reach a non-zero plateau); the transfer to ν^{op} is open (§ 2.2, the DPI direction is reversed; § 7 point iii). The thesis "inevitability of $\eta_L \to 0$" rests on the *budget* channel, the thesis "growth of nostalgia ν" on the *theoretical* normalisation; both are consistent (under polynomial memory both act, under exponential memory the budget one acts, even when ν is held below 1).

5. The role of the refresh protocol. The conclusion depends on how the refresh budget is distributed over bits. Under FIFO the fresh fraction $\phi(t) \to 0$ under polynomial growth, and the undertow saturates. Under *proportional* refresh the refresh period of a bit $|M_{\leq t}|/\dot{M}_{\text{refresh}}$ also grows, and the fresh fraction decays (the same conclusion $\nu \to 1$). Escape requires a bounded refresh period for each bit, which at finite $\dot{M}_{\text{refresh}}$ is incompatible with unbounded capacity growth except at the cost of a growing E_{actual}. Thus the thermodynamic cost arises in *any* protocol that distributes refresh over raw bits: through E_{grow} (exponential capacity, S3.4) or E_{actual}. Separately, the *recursively-compressing* protocol (Kalman type): an $O(1)$ sufficient statistic, $|M_{\leq t}| = O(1)$, $E_{\text{grow}} = 0$. It too yields no escape, but for an *informational* rather than a cost reason: the compressing filter realises exactly the tracking floor of C1 ($n^{\star}_{\text{eff}} = (1 - g^{\star})/g^{\star}$, § 3) and the corresponding non-zero nostalgia — the $O(1)$ statistic has already extracted all the information about $\theta(t)$, and the old observations are decorrelated (the mechanism $\beta(\Delta t) \to 0$ of Theorem 2). Bounded memory does not save the situation because it is cheap, but because it runs informationally into the floor of C1 (in the window A1; the exact softmax model is open, § 7 point iv). Separation of the carriers of inevitability: under *compression/recycling* ($|M_{\leq t}| = O(1)$, $E_{\text{grow}} = 0$) inevitability is carried by the *informational floor of C1*, not by the cost of C3; under *growth of raw material* ($|M_{\leq t}| \to \infty$) the C3 cost diverges (with the caveat A1).

5a. The logical status of C3 relative to C2. To avoid overclaiming independence, we localise what in Theorem 3 is *derivable* from Theorem 2 and what is genuinely new. **The saturating part of C3** ($\nu^{\text{theor}} \to 1$ under polynomial growth, S3.2) is *not independent* of C2: it is a *pointwise application* of the DPI mechanism of Theorem 2 (step S2.0) with the constant c replaced by $\phi(t)$ — it inherits the entire mixing class of C2 (4.1) and the bounds ($\int \beta < \infty$; fOU $H < 3/4$). "No finite polynomial threshold α_c" in this part is an *expansion identity* $\phi = 1 - (1 - \tau_E/t)^\alpha = \alpha\tau_E/t\,(1 + o(1)) \to 0$ for any finite α (leading $\gamma \equiv 1$), not a separate mechanism. Hence in the saturation part C3 is *closed off by the same mixing-induced floor that carries C2.* **The genuinely new content of C3, not derivable from C2**, is three points: (i) the *adiabatic carry-over* of the S2.0 majorant (derived for *constant* c) to a *slowly-varying* $\phi(t)$ with $O(\tau_E/t)$ control of non-constancy (§ S3.2) — C2 has none of this; (ii) the *budget channel of cost divergence* $E_{\text{grow}} + E_{\text{store}} \propto e^{\kappa t}$ (point 4, § S3.4) — C2 has no notion of the cost of capacity growth; (iii) the *phase boundary* polynomial/exponential as a direct answer to § 8.5 #2 [3]. Accordingly, the three routes are closed off not by three *independent* mechanisms, but by *two*: the mixing-induced floor (carrying C2 and the saturating part of C3) and the budget channel of growth cost (the novelty of C3); C1 is closed onto the same mixing through the tracking floor (§ 8). This is consistent with § 1 and § 8 (C1$\leftrightarrow$C2, C1$\leftrightarrow$C3); here the containment C3-saturation$\subset$C2 is added.

6. Interpretation. Theorem 3 closes the picture: memory growth offers no refuge — polynomial expansion does not change the asymptotics of nostalgia, and exponential growth is paid for by a divergent E_{grow}. The rational alternative to escape is *reset* of the stale layer ($\nu \to 0$ at the cost of a one-off payment, cheaper than continuous expansion): the optimal τ^*_{reset} [3] (12), biologically — epigenetic reprogramming, diapause (§ 8 [3]). In machine learning the parallel runs along the axis of *stored* memory (context, retrieval index, replay buffer), not the number of parameters: factual knowledge becomes stale with corpus shift [26, 27], and non-stationary streams require continual refresh, not mere expansion [28].

6. Numerical verification

This section records the results of the numerical verification of the three theorems. The simulations inherit the apparatus of § 6 [3] (Markov chains with logit drift), are implemented in `simulations/{ou_adiab,fou_floor,psp_floor,superlinear_memory}/` of the mirror repository (numpy, fixed seeds, `results_summary.{txt,json}`) and were actually run; the code is not reproduced here. A methodological separation: the proofs are carried by § S1–S3, and the simulations provide independent numerical support but do not replace them [3] § S5.2.

1. ***`ou_adiab`*** *— is the convergence $B \to K/2$ realizable under OU drift? (Theorem 1).* A direct test of the realizability of the coefficient **in the exact linear-Gaussian (Kalman) setting of Theorem 1** (A1 is *exact* here), which separates the property of the BNT logarithm from artefacts of the softmax learner. Model: $K = 56$ independent scalar OU coordinates (2.1) with Gaussian observations and an **optimal Kalman filter**; the noise r sets $I_{\text{rate}} = 1/r$ and $\epsilon_{\text{track}} = \sigma^2/(\lambda^2 r)$. $k = 8$, $\lambda = 10^{-2}$, $P_0 = 10^6$, 40 realizations, seed 20260527, bootstrap CI on B. Three experiments: (A) static anchor without drift (L_{excess} must be $(K/2)\ln t$); (B) local log-slope under drift; (C) a scan over ϵ_{track} in a *wide* window $[3\tau_{\text{conc}},\, 0.1/\sigma^2]$ ($t \gg \tau_{\text{conc}}$, fit *after* the spike; $W \approx 150$).

Actual run result. **(A)** the static slope slope$/(K/2) = 0.990$ ($R^2 = 0.99978$) — the BNT logarithm is real without drift, the pipeline is correct. **(B)** under drift the log-slope equals $K/2$ only in the short prior-collapse burst ($t \lesssim \tau_{\text{conc}}$): slope 0.936 on $[1, 20]$; thereafter $g^\star$ bounds $n_{\text{eff}} \approx 1/g^\star$, the log component levels off onto a **plateau** $\sim (K/2)\ln(1/g^\star)$, and the slope of the log component in the window $[100,\, 0.5\tau_{\text{conc}}]$ falls to 0.099 (the full L_{excess} continues its linear growth $A \cdot t$ — not a global collapse, but saturation of the log term). **(C)** in the wide window $B/(K/2)$ does not approach 1 as $\epsilon_{\text{track}} \to 0$: across 10 logarithmically uniform values of ϵ_{track} from $3 \cdot 10^{-1}$ to $3 \cdot 10^{-2}$ it fluctuates in $[-0.31;\, 0.16]$ around zero (9 of 10 bootstrap CIs cover 0; the single exception lies on the *negative* side — even further from the target $K/2$, i.e. the statistical outlier expected among 10 independent 95% CIs, not a sign of convergence), $R^2 = 1.000$.

Key observation of the scan: across all 10 points (lowering ϵ_{track} by an order of magnitude) the width of the log-carrying window stays constant — $W \in [156,\, 166] \approx$ const (Fig. 1). This is direct numerical confirmation of the σ-invariance of the window $W \asymp \mathcal{I}_{\text{rate}}/\lambda = 1/(\lambda r)$ (Proposition 1, § 3): the limit $\epsilon_{\text{track}} \to 0$ does *not* open the window — the strongest evidence of the structural unattainability of the slope. The working hypothesis "push $\epsilon_{\text{track}} \to 0$" is refuted: the window in which the logarithm lives ($t \lesssim \tau_{\text{conc}}$) is structurally excluded by the requirement $t \gg \tau_{\text{conc}}$ (τ_{conc} and the ceiling both $\propto 1/\sigma^2$). **Conclusion:** the identity $B = K/2$ is real, but is realized *only transiently*; numerical convergence as a slope in a wide window is **unattainable for a learner with saturating memory** (growing memory does not save it — the floor is a lower bound across all architectures, § 3, Proposition 1, § S1.6). Details — § S4.1. *Falsifier:* static slope $\neq K/2$ (not observed — 0.990); recovery $B/(K/2) \to 1$ in a wide window (not observed — the series is flat near 0).

2. `fou_floor` — the nostalgia floor and the rate exponent for fractional OU (Theorem 2). A real trajectory simulation on **exact stationary fOU** (not a recomputation of a closed-form formula): $K = 56$ independent stationary fOU trajectories with $H \in \{0.3,\ 0.5,\ 0.7,\ 0.9\}$ are generated by the **spectral-circulant method** from the density $S(\omega) \propto |\omega|^{1-2H}/(\lambda^2 + \omega^2)$; the eigenvalues of the Davies–Harte circulant are non-negative by construction (zero-truncation), and the target autocovariance $\gamma(k)$ is the inverse DFT of these eigenvalues. **Covariance control** confirms $\hat\rho(s) \approx \gamma(s)$ (relative error $< 5\%$ up to lag $\sim$ 200–400). FIFO memory with $\tau_E = 200$, $c \approx 0.3$, $|M| = 400$; a bit is predictive as long as $\hat\rho^2(\text{age}) \geq \hat\rho^2(\tau_E)$.

Actual run result (seed 20260527, 40 realizations, path $3.3 \cdot 10^4$, circulant $M = 6.6 \cdot 10^4$): for all H — the same floor $\liminf \nu = 0.700 \approx 1 - c$ (spread across H exactly 0.000; Fig. S2). An honest caveat on the *status of the evidence* (an asymmetry with `psp_floor`): this floor is *by construction* — the deterministic age threshold $\hat\rho^2(\text{age}) \geq \hat\rho^2(\tau_E)$ pins the fresh fraction to c, hence the floor to $1-c$ identically; the zero spread is a consequence of the pinning, not an independent measurement (the proof is carried by § S2; § S4.5). The independent content of `fou_floor` lies *not* in the value of

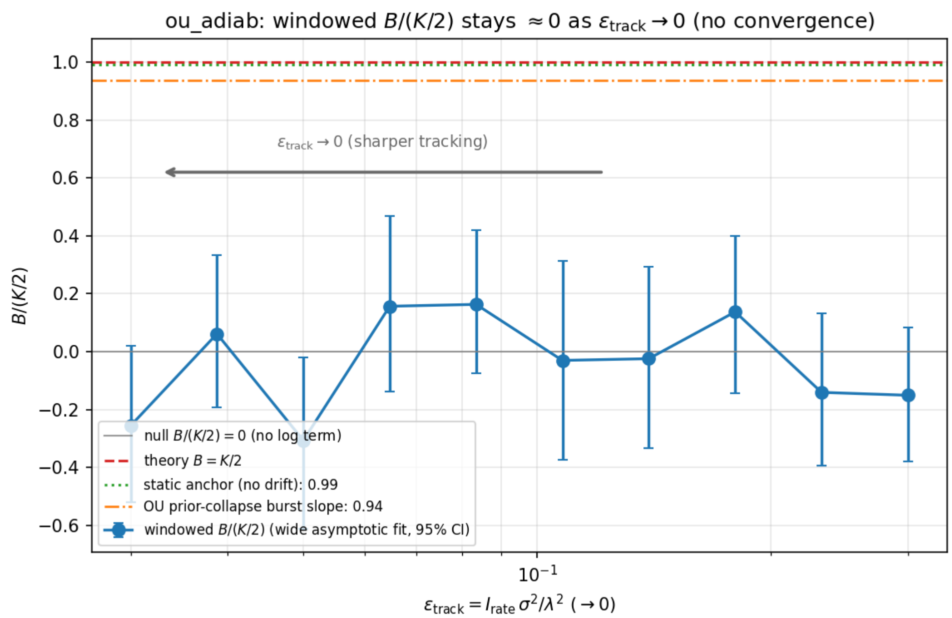

Figure 1. Structural unattainability of the slope $K/2$ (C1, Proposition 1; series `ou_adiab`). The ratio $B/(K/2)$, estimated in the wide asymptotic window, stays flat near zero as ϵ_{track} is lowered by an order of magnitude — rather than rising toward 1 — whereas the static anchor (without drift) yields $K/2$ (0.990), and the transient prior-collapse burst yields 0.936. The log-carrying window is σ-invariant ($W \approx 156$–$166 \approx$ const across all 10 points; the values of W are reported in § 6.1, not plotted on the graph), so the limit $\epsilon_{\text{track}} \to 0$ does not open it. The grey line is the null hypothesis $B/(K/2) = 0$ (no log term); all points sit on it.

the floor, but in (i) the covariance control (the paths are genuine stationary fOU, not a surrogate) and (ii) the confirmation of the *exponent* $4(1-H)$ in the far tail.

The rate exponent confirms the analytical $4(1-H)$ (Fig. S4) — on a correct process, in the far tail, by the exact covariance γ (the sample ACF on long memory is biased toward zero at large lags). In the far window ($\Delta \gtrsim 8\tau_E$) the local log-log slope of ρ^2 matches $4(1-H)$, and the two values of H play different evidential roles: $H = 0.7$ ($H < 3/4$, $\int\beta$ finite) — *inside* the strict Theorem 2, with the match 1.237 on $[8, 40]\tau_E$ and 1.201 on $[20, 80]\tau_E$ (analytics 1.2) — an *in-theorem confirmation*; $H = 0.9$ ($H \geq 3/4$, $\int\beta$ divergent) — *outside* the theorem, with the match 0.406 and 0.385 (analytics 0.4) — only a *numerically supported hypothesis*. In the near window $[\tau_E, 8\tau_E]$ the exponent is not isolated (1.62 at $H = 0.7$, 0.46 at $H = 0.9$ by the exact covariance — the OU "knee"; 2.44 and 1.40 by the sample ACF — estimator bias). **This is an upgrade of C2:** the previous surrogate AR(1)$\circ$fGn gave 2.23 and 1.26, outside the CI of $4(1-H)$ — a property of the surrogate, not of the theory; on correct fOU both the depth $(1-c)$ and the exponent are reproduced. The qualitative dichotomy holds: at $H = 0.5$ the tail is exponential (rate $4.07 \cdot 10^{-3}$/step), at $H > 0.5$ — power-law. *Falsifier:* divergence of the floor *across* H or a far-tail exponent outside $4(1-H)$ (not observed). The load-bearing contribution of C2 — the *depth* $(1-c)$ — is reproduced

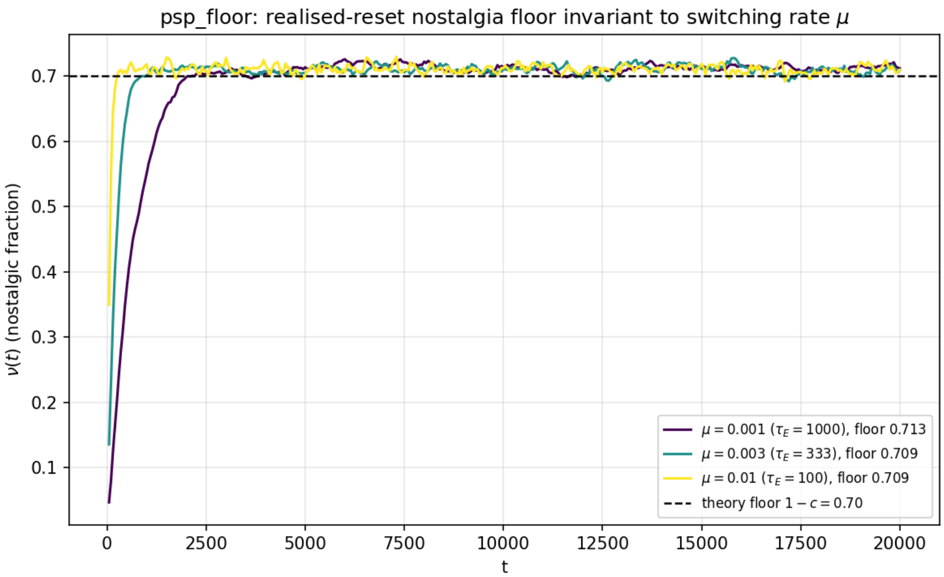

Figure 2. Invariance of the depth of the nostalgia floor across the refresh rate (C2, Theorem 2; series `psp_floor` — *independent* evidence). The steady-state floor $\nu \approx 0.709$–$0.713 \approx 1 - c$ is the same for the Poisson-reset intensity μ over a 10-fold range ($\tau_E = 1/\mu \in \{1000, 333, 100\}$), even though the *approach dynamics* toward the floor differ with μ (three distinct curves are visible). Unlike the by-construction floor of `fou_floor` (Fig. S2), here the reset is a *realized* Poisson stream, so the floor lies slightly above the theoretical 0.70 — an independent measurement of the depth. The depth is set by the refresh rate c, not by the drift frequency.

invariantly across H; the exponent is secondary and asymptotic (§ 7 item vi).

3. ***`psp_floor`*** *— the nostalgia floor for Poisson reset (Theorem 2).* A real trajectory simulation of the PSP surrogate of § 6.1 [3]: each coordinate carries a *realized* Poisson stream of resets of intensity $\mu = 1/\tau_E$; a retained bit is predictive as long as its coordinate has not actually reset. The floor and the decorrelation rate are read off the realized process, not from the formula $e^{-\mu\Delta}$. *Actual run result (seed 20260528, 40 realizations,* $|M| = 400$*,* $c \approx 0.3$*,* $\mu \in \{10^{-3}, 3 \cdot 10^{-3}, 10^{-2}\}$*):* the steady-state empirical floor $\hat{\nu}_{\text{floor}} = 0.709$–$0.713 \approx 1 - c$ over a 10-fold range of μ (spread 0.004; Fig. 2) — invariance to the switching frequency is confirmed. A substantive asymmetry with `fou_floor`: here the reset is a *realized* stream, not a hard-wired threshold, so the floor 0.709–0.713 is *independent* evidence (systematically slightly above the theoretical 0.70); `psp_floor` carries more independent content than the by-construction `fou_floor`. The decorrelation rate from the realized survival curve $\hat{S}(\text{age})$ equals $1.01 \cdot 10^{-3}$, $3.01 \cdot 10^{-3}$, $1.00 \cdot 10^{-2}$, and each bootstrap CI covers the nominal μ — a genuine measurement of the scale $1/\mu$. Jointly, `psp_floor` (exponential *rate* μ) and `fou_floor` (power-law *exponent* $4(1 - H)$) exercise both ends of the mixing class. *Falsifier:* dependence of the floor on μ at fixed c; a decorrelation rate outside the CI of μ (not observed).

4. ***`superlinear_memory`*** *— the phase boundary (Theorem 3).* The memory grows

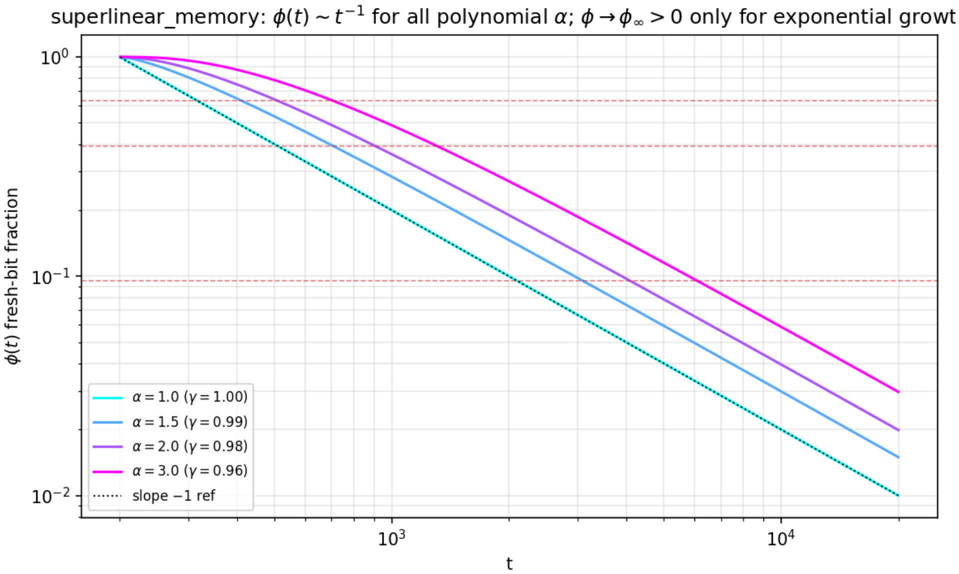

Figure 3. Absence of a finite polynomial threshold α_c (C3, Theorem 3; series `superlinear_memory`). The fresh-bit fraction $\phi(t) \sim t^{-\gamma}$ has leading exponent $\gamma \equiv 1$ for all polynomial $\alpha \in \{1; 1.5; 2; 3\}$ (prefactor $\phi \cdot t \to \alpha\tau_E$), whence $\nu(t) \to 1$ — the undertow *saturates* at any finite α. The apparent downward drift of γ with growing α on the original window is a finite-window subleading artefact of the exact $\phi = 1 - ((t - \tau_E)/t)^\alpha$, removed by widening the fit window (convergence diagnostic, § S4.4). Escape is possible only through exponential capacity growth — Fig. S6.

as $|M_{\leq t}| \propto t^\alpha$ for $\alpha \in \{1,\ 1.5,\ 2,\ 3\}$ and, separately, exponentially as $e^{\kappa t}$; $\nu(t)$ and the fresh fraction $\phi(t)$ are measured. *Actual run result (seed 20260529, $\tau_E = 200$):* for all polynomial α the fresh fraction $\phi(t) \sim t^{-\gamma}$ with $\gamma \in \{1.000; 0.991; 0.982; 0.964\}$ on the baseline window, and the prefactor $\phi \cdot t \to \alpha\tau_E$ (200, 298, 394, 581 against 200, 300, 400, 600), whence $\nu(t) \to 1$: there is no polynomial α_c (Fig. 3). The leading exponent is identically $\gamma = 1$; the downward drift of γ with growing α is finite-window subleading curvature in $\phi = 1 - ((t - \tau_E)/t)^\alpha$, removed by widening the window: $\times 10$ gives $\gamma \to \{1.000; 0.999; 0.998; 0.996\}$, and $\times 1000$ — all 1.000 (diagnostic `gamma_window_convergence`, § S4.4). For exponential growth ($\kappa\tau_E \in \{0.1; 0.5; 1\}$) — $\phi_\infty = 1 - e^{-\kappa\tau_E} \in \{0.095; 0.394; 0.632\} > 0$ (nostalgia below 1), but $E_{\text{grow}}(t) \propto e^{\kappa t}$ diverges (the ratio $E_{\text{grow}}(T)/E_{\text{grow}}(0)$ reaches up to $9.8 \cdot 10^{42}$), and $\eta_L(t) \to 0$ through the denominator (5.3). *Falsifier:* $\phi(t)$ decaying other than as t^{-1} under polynomial growth, or saturation of $\phi(t)$ at finite α, refutes the absence of α_c.

5. *What the simulations do not do.* (i) They do *not* prove the theorems — they are illustrations; the strict statements are carried by § S1–S3. (ii) They work only with a known true $\theta(t)$, so $\nu^{\text{theor}} = \nu^{\text{op}}$ by construction [3] § 6; the operational indistinguishability of the two normalizations is not tested. (iii) They do not go beyond abstract Markov chains with logit drift. (iv) The convergence $B \to K/2$ is tested on a finite scan — support, not proof. (v) The class of non-mixing (heavy-tailed)

environments is not simulated — it lies outside the scope of Theorem 2.

7. Discussion

1. Relation to non-stationary online learning. The cumulative excess loss $L_{\text{excess}}(t)$ is the dynamic regret of the learner against an oracle that tracks the drifting optimum, and Theorem 1 has a direct parallel in this theory. Two budgets of change: (i) the *path-length* $P_T = \sum_t \|\theta(t) - \theta(t-1)\|$, introduced by Zinkevich [19], who proved the dynamic-regret bound $O(\sqrt{T}\,(1+P_T))$ against a drifting comparator (and $O(\sqrt{T})$ against a static one); Hall–Willett [29] carried it over to dynamic mirror descent in drifting environments, and the sharp form $O(\sqrt{T(1+P_T)})$ with a matching lower bound was given by Zhang–Lu–Zhou [30]; (ii) the *variation budget* V_T in the minimax theory of Besbes–Gur–Zeevi [31], giving $\Theta(T^{2/3}V_T^{1/3})$. For smooth OU drift with stationary increments both budgets grow linearly ($P_T, V_T \propto T$): the BGZ minimax bound then gives exactly *linear* regret $\Theta(T)$, and the Hall–Willett tracking bound gives a superlinear upper estimate $O(T^{3/2})$ that majorizes it. In both cases this is consistent with the linear term $A \cdot t$ of our decomposition (3.1): BGZ reproduces it as the leading term, Hall–Willett majorizes it. The refinement of Theorem 1 concerns not the leading $\Theta(T)$ term, but the *sub-linear* logarithmic correction $(K/2)\ln(\lambda t)$ on top of it — and it is precisely this that saturates into the plateau (Proposition 1). *Logarithmic* regret in this line is achieved not by Zinkevich ($\sqrt{T}$), but, for strongly convex losses, by Hazan's algorithms [15] and in drifting environments [20]; our $(K/2)\ln$ asymptotics is closer to this class. The difference of forms (power-law BGZ versus logarithmic) reflects the difference of formulations: BGZ bound the *total* variation, we bound the *spectral* structure of the drift. The role of mixing coincides structurally (the optimal forgetting scale τ_f^{forget} is tuned to τ_E), as does monotonicity in τ_E [3] § S5.4. The connection of dynamic-regret bounds [32, 15] to η_L through the identity "regret = excess-loss = unpaid predictive value" deserves a separate study.

1a. The necessity of narrowing the class. Theorem 2 extends the domain of the floor, but does not remove the fundamental limitation: without an a priori restriction on the drift class, sublinear dynamic regret is unattainable — under an unbounded variation budget every learner incurs regret no slower than linear [31, 19]. The mixing condition (4.1) is a necessary narrowing: it selects environments that forget their past; outside it (heavy-tailed, non-recurrent drift) neither the floor $1 - c$ nor the adiabatic asymptotics is guaranteed. The boundary of the admissible class of § 4 is not a defect but a reflection of this impossibility: the undertow is inevitable exactly for mixing environments, and this is the maximal class on which the statement holds. In its philosophical background this is in the spirit of no-free-lunch [33], but formally we do not rely on NFL (the load-bearing argument is the impossibility of sublinear dynamic regret); moreover, [33] is an NFL of *supervised learning*, averaging over classes of target functions rather than over *environments* with a fixed spectral structure, so it applies only as an analogy.

1b. Demarcation from related notions. To avoid rebranding, we separate three notions. *Nostalgia* — the retention of a stale memory layer that no longer predicts anything but *continues to be paid for* thermodynamically. *Catastrophic forgetting* [34, 35] — the opposite pathology: the *loss* of still-useful memory during retraining. *Concept drift* [28] — a property of the *environment* itself (a distribution shift), a cause rather than a consequence. Nostalgia does not reduce to any of these: it is the thermodynamic cost of *successful* retention (unlike forgetting) under drift (concept drift as an external cause), and its subject is the unpaidness of retention, not the preservation or loss of information.

2. Kalman tracking. The EWMA filter of Theorem 1 is a special case of the stationary Kalman–Bucy filter for a linear-Gaussian system with a hidden OU state [16, 17]; the stationary error $\Sigma_\infty \propto \sigma^2/\lambda$ (the root of the algebraic Riccati equation) is carried over into the thermodynamic context as the linear coefficient A. The nonzero *steady-state tracking floor* itself is the classical *lag misadjustment* of adaptive filtering [36, 37]. Our contribution is not the existence of the floor, but its thermodynamic reading (floor $\Leftrightarrow$ linear term A and the plateau of the logarithm) and its role as an information-theoretic lower bound on n_{eff} (§ S1.6). The generalization to the nonlinear softmax model (extended/unscented Kalman) is a technical extension, parallel to Remark 6 [3].

3. Renewal theory. The PSP class of Theorem 2 is a renewal process: the reset points form a Poisson stream, and the ages of bits are residual lifetimes. The uniformity of the age distribution under FIFO (§ 4.2, § S2.0) is a consequence of the *stationary age distribution* of a fixed-size buffer (the inspection paradox / backward recurrence time, [38] ch. 5), and *not* of the elementary renewal theorem. The power-law mixing of fOU at $H > 1/2$ corresponds to heavy-tailed renewal — a natural continuation.

4. What remains open. (i) *Heavy-tailed and non-Markov drift without mixing*: Theorem 2 requires $\beta(s) \to 0$; environments with divergent autocorrelation fall out of the class, and the floor $1 - c$ is not guaranteed for them. (ii) *A bridge to fluctuation theorems*: the decomposition of the budget into the flows $E_{\text{store}}, E_{\text{grow}}, E_{\text{actual}}$ [3] (1) and the Crooks/Jarzynski theorems with feedback control [7], the inequalities for bipartite systems and transfer entropy [39], and the stochastic thermodynamics of computation [40] (for a general review of information thermodynamics, see [41]) may yield finer inequalities for each flow — a direction for future work. (iii) *Operational versus theoretical efficiency*: Theorem 1 is proved for the counterfactual η_L^{excess}; the carry-over to the measurable $\eta_L(t)$ remains structural. (iv) *A strict transient filter*: the derivation of $K/2$ rests on the stationary Riccati and the heuristic $n_{\text{eff}} \asymp \lambda t$ (A3); a strict non-stationary Riccati and control of the softmax-linearization residual (A1) beyond the adiabatic window are open. (v) *Numerical asymptotic realizability of $B \to K/2$ (established as unattainable)*: the extended test with an *optimal* Kalman learner (§ 6.1, § S4.1) refuted the previous internal working hypothesis — the attempt to push $\epsilon_{\text{track}} \to 0$ in order to recover $\hat{B} \to K/2$ as a slope (it is the attempt that is refuted, *not* the hypothesis of § S8.1 #2 [3], which only conjectured the functional form and is analytically confirmed by Theorem 1). Under genuine OU drift n_{eff} is bounded by the floor, L_{excess} levels off onto a plateau, and in a wide window $B/(K/2) \to 0$ (§ 6.1); the obstruction is structural (τ_{conc} and the adiabatic ceiling both $\propto 1/\sigma^2$). Growing memory does not save it: the floor is the MMSE lower bound on n_{eff} across all memory architectures (§ S1.6, § 3, Proposition 1). Hence the question of growing memory is neither "open" nor "closed onto C3", but *removed* by the optimality of the floor in the window A1 (accumulating raw material is both fruitless and costly — the regime of Theorem 3; the non-Gaussian model is open, item iv). (vi) *The empirical undertow-rate exponent of fOU*: the load-bearing contribution of C2 — the *depth* of the floor $(1-c)$ — is proved and does not depend on the rate exponent; the *exponent* $4(1-H)$ is secondary and *asymptotic* — on exact stationary fOU it is confirmed in the far tail ($\Delta \gtrsim 8\tau_E$: 1.20 at $H = 0.7$ against analytics 1.2; 0.39–0.41 at $H = 0.9$ against 0.4), but in the near window it is not isolated (the OU "knee", the biased sample ACF), and is read off the exact covariance (§ 6.2, § S4.2). Structurally this is *the same situation as the transience in C1*; on correct fOU (not the surrogate AR(1)∘fGn) both the depth $(1-c)$ and the exponent are reproduced, which **strengthens** C2. Open is only the strict control of the convergence rate at $H \geq 3/4$ — and this does not undermine the floor.

8. Relation to companion papers #1 and #2

The present work is the third step of the programme of Landauer efficiency of self-modeling, and it extends, rather than revises, both predecessors.

Summary without formulae. The undertow is inevitable for one simple reason: all three ways of escaping it run into the same thing. Old knowledge inevitably ceases to correlate with the present — the environment forgets its past, and with it the memory recorded about that past depreciates; yet one must pay ever more to store this memory. Therefore neither more accurate tracking of the environment, nor faster updating of memory, nor expansion of its capacity provides escape: the three escape routes are closed off, and behind them stand only two mechanisms — old memory decorrelates, and the cost of storing it grows. The technical analysis of this closure follows below.

The inevitability of the undertow, in summary. The three theorems close off the three escape routes, all three dead ends (accurate tracking — the no-go of realizability of Proposition 1; memory updating — the universal floor $1-c$ of Theorem 2; capacity growth — the absence of a polynomial threshold of Theorem 3), and jointly $\eta_L \to 0$ inevitably. The level qualifier (from § 7 item iv): the no-go is conditional on the postulates of self-payment and stationary accounting (A1)+(A3) [3] and on the linear-Gaussian adiabatic approximation; "inevitability" is proved *in the adiabatic window / for mixing environments* ($\beta(\Delta t) \to 0$, § 2.6, § 4) — the strict version beyond the class (non-adiabatic, heavy-tailed drift) is open (§ 4.6, § 7 items i, iv). Within these limits the obsolescence of non-stationary memory is neither an artefact nor a consequence of suboptimality, but a structural property: the rational response is not to increase accuracy or capacity, but periodically to *reset* the stale layer (the optimal τ^*_{reset} [3] (12), § 5 item 6).

Interlocking of the routes. The three routes are closed off not *independently*, but *interlockingly*: an attempt to escape one is either fruitless already at the information level, or runs into another. This removes the objection that the inevitability is conditional — "only for saturating memory", whereas a learner with *growing* memory would supposedly raise n_{eff} relative to $\theta(t)$ above the floor $(1-g^\star)/g^\star$. The objection is wrong already information-theoretically: the floor is the *optimum of tracking* — a lower bound on n_{eff} across all memory architectures, not the limit of a particular capacity (§ 7 item v; § S1.6). Therefore the attempt to raise n_{eff} is either (i) information-theoretically fruitless (the optimal filter is already at the floor — C1, the same decorrelation mechanism $\beta(\Delta t) \to 0$ that carries the floor of C2)‡, or (ii) suboptimal by accumulating raw material — fruitless *and* costly (the regime of Theorem 3: $E_{\text{grow}}, E_{\text{store}} \propto e^{\kappa t}$, $\eta_L \to 0$ through the denominator (5.3)–(5.4)). Growing memory does *not* open a fourth route, and the "inevitability" is **not conditional** (in the linear-Gaussian window A1; the non-Gaussian model is open, § 7 item iv). The routes are three facets of one closed loop; the load-bearing mechanisms are *two* (the mixing-induced floor and the budget channel, expanded below), with C1 closed onto mixing through the tracking floor.

What is strictly new as a whole. Each individual brick is honestly attributed (§ 3 item 6 — the floor as BNT class II; § 4 item 3 — the DPI mechanism of the floor; § 5 item 5a — the saturating part $\subset$ C2), so the independent contribution of the work is *not* in any single result, but in the *closure of the loop*. Three a-priori-independent routes reduce to *two* load-bearing mechanisms, both closed off on top of the Landauer denominator: (i) the *mixing-induced floor* — the decorrelation that sets both the nostalgia floor (C2) and the tracking floor (C1), and that carries, through the pointwise S2.0, the saturating part of C3 (§ 5 item 5a); (ii) the *budget channel* — the divergent cost of capacity growth $E_{\text{grow}} + E_{\text{store}} \propto e^{\kappa t}$ (the load-bearing novelty of C3, § 5 item 4), which C2 lacks. This is not three positive results, but a *closure theorem* of the escape routes (a no-go of synthesis), and the closure itself is a testable prediction — the vanishing $\hat{B} \to 0$ in a wide window (§ 6), which is confirmed. Beyond companion paper #2, three things are strictly new: (i) the DPI generalization of the floor from the OU spectral gap to general mixing (capturing the non-Markov fOU through a multiple, rather than pairwise, correlation — § 4); (ii) the budget channel of the divergence of the cost of capacity growth, absent from #2 (§ 5 item 4); (iii) the demonstration that the three routes are *not* independent — and this, rather than any one of them on its own, is the structural result of the work.

Nor does a fourth route arise from the quasi-stationary limit of very slow drift $\lambda \to 0$, in which $n_{\text{eff}} \sim t$ could grow ($\tau_E = 1/\lambda \to \infty$ carries the obsolescence horizon off to infinity). With adiabaticity preserved, this limit *continuously degenerates into the stationary regime*, where there is no undertow *by definition* rather than its being overcome: the growth $n_{\text{eff}} \sim t$ is the disappearance of the drift itself, and the floor

‡ The floor speaks of decorrelation from the *current* $\theta(t)$, whereas the mechanism of Theorem 2 (4.1) speaks of decorrelation from the *future trajectory* $\theta_{[t,t+\tau]}$; for a horizon $\tau = O(1)$ both have the same decay exponent (for fOU the trajectory supremum is attained at the near end, preserving the leading $4(1-H)$). The full argument — § S2.0, Remark "trajectory versus pair".

$1/g^\star \to \infty$, together with $g^\star \propto \sigma^2 \to 0$, ceases to be a constraint. Neither finite λ (the floor is active) nor the limit $\lambda \to 0$ (the drift disappears) opens an escape.

Relation to companion paper #2 [3]. All three theorems advance the open questions of § 8.5 [3] without coming into conflict with it. Theorem 1 *resolves* aspect (i) of the conjecture of § S8.1 *negatively*: the value $B = K/2$ is confirmed in the limit $\epsilon_{\text{track}} \to 0$ *as transient* (realizable only on the window $t \lesssim \tau_{\text{conc}}$), but no attainable asymptotic realization of the slope exists under drift (a no-go, Proposition 1) — the resolution of an open question, not a refutation; the carry-over to the operational η_L is open (§ 6.1, § 7). Theorem 2 *embeds* Lemma 2 as the special case $H = 1/2$: the OU spectral gap is a sufficient but not a necessary condition for the floor; the necessary one is mixing (4.1). Theorem 3 closes the question of superlinear memory: escape is impossible polynomially and thermodynamically costly exponentially. The apparatus of [3] (η_L, $N_{\max}$, ν, OU parametrization, decomposition) is used without redefinition.

Relation to companion paper #1 [42]. The stationary lemma [42] § 2.1 is a special case of the non-stationary budget at $\dot{E}_{\text{store}} = \dot{E}_{\text{grow}} = 0$ [3] § 2.2. In the stationary limit of zero drift the *drift-induced* nostalgia $\nu \to 0$ and L_{excess} does not grow logarithmically — both theorems degenerate into the stationary picture. To avoid conflating mechanisms: this $\nu \to 0$ does *not* cancel the "nostalgia bits" of lemma #1, which arise from a different mechanism — the sink of finite memory (retention beyond the correlation scale even without drift), nonzero also in the stationary limit. The mechanisms are orthogonal: the drift-induced undertow vanishes at zero drift, the finite-memory sink does not.

Progressiveness of the extension. The programme satisfies the Lakatosian criterion of progressiveness: each work adds excess, independently testable content. Paper #1 — the stationary scale; paper #2 — the non-stationary extension with three open questions; the present work — three theorems reducing the three escape routes to a single thesis of inevitability. The negative result (Proposition 1) is a *progressive*, not a degenerative, shift: it does not rescue the conjecture of § S8.1 by an auxiliary patch, but makes a testable prediction ($\hat{B} \to 0$ in a wide window, § 6), which is confirmed. The work generates new open problems (the convergence rate to $4(1-H)$ at $H \geq 3/4$; heavy-tailed drift; a bridge to fluctuation theorems) for future works. Operationalization with *growing* memory does *not* remain an open problem here — it is closed by the optimality of the floor, forming no independent route (§ 7 item v).

Substrate-independence. The undertow is a property of the information thermodynamics of non-stationary memory, not of a particular substrate: neither the ceiling of the numerator ($I_{\text{pred}} \leq \ln k$) nor the divergence of the denominator (the Landauer budget) depends on whether the memory is stored by synapses, network weights, or an epigenetic mark. Hence a unified reading of why adaptive systems on different substrates converge to one strategy — periodically to *reset* the stale layer rather than accumulate memory without limit (continual learning with replay, synaptic turnover, evolutionary bet-hedging — § 5 item 6, § 7 item 1b): candidates for the same no-go. This is a parallel, not a corollary — the theorems are proved only in their model class (mixing slow drift, the adiabatic window A1), and verification on concrete systems remains a separate task.

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Author Contributions. Sole author (A.A.): conceptualization, proofs, numerical design, and writing.

Data Availability. The code of the numerical simulations (§ 6) and the source files of the work are openly available in the public mirror repository `andriishin/landauer-undertow-oa` (GitHub) and archived on Zenodo (DOI 10.5281/zenodo.20836705). The work uses no external empirical data (a theoretical work with numerical illustrations on abstract models).

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